

A System-Level Security–Utility Evaluation Framework for Privacy-Preserving Multimodal Biometric Verification Using Cancellable Templates and CKKS Homomorphic Encryption

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ABSTRACT

Biometric authentication is widely used for identity verification because biometric traits such as face, fingerprint, and iris are strongly associated with an individual and are difficult to forget, share, or lose. However, biometric templates are sensitive, permanent, and difficult to revoke after compromise. If a stored biometric representation is exposed, the underlying biometric trait cannot be replaced in the same way as a password. This creates serious privacy risks, including unauthorized matching, template misuse, and cross-application linkage. These risks become more significant in multimodal biometric systems because multiple identity-bearing traits are processed together.

This Major Project-II presents a system-level security–utility evaluation framework for privacy-preserving multimodal biometric verification using face, fingerprint, and iris modalities. The work extends the Major Project-I baseline, where plain-domain matching, CKKS encrypted matching, and score-level fusion were evaluated. In the present phase, the framework is strengthened by adding cancellable biometric template transformation, same-key and cross-key evaluation, protected score-level fusion, and CKKS encrypted matching over protected templates. The work does not propose a new biometric recognition model, a new cancellable transformation algorithm, or a new cryptographic scheme. Instead, it establishes how template-level protection, multimodal fusion, unlinkability evaluation, encrypted-domain matching, and runtime overhead interact under one controlled verification protocol.

A virtually subject-aligned multimodal dataset was constructed using VGGFace2 for face, SOCOFing-real for fingerprint, and CASIA-Iris-Thousand for iris. The dataset contains 600 virtual subjects, with 10 samples per modality per subject, giving 18,000 biometric samples in total. A subject-level split was used with 480 training subjects, 60 validation subjects, and 60 testing subjects. For validation and testing, balanced verification-pair sets were generated with 5,400 pairs per split, consisting of 2,700 genuine and 2,700 impostor comparisons. Deep

feature embeddings of 512 dimensions were extracted for each modality. Face embeddings were obtained using InceptionResnetV1 pretrained on VGGFace2, fingerprint embeddings using a ResNet-based model, and iris embeddings using an EfficientNet-B3-based model.

For template protection, cancellable transformation was implemented using key-dependent random projection. Protected templates were evaluated using projection dimensions of 64, 128, 256, 512, and 1024, with both binary and non-binary variants. Same-key evaluation was used to measure protected-template utility, while cross-key evaluation was used to analyse practical unlinkability and renewability. CKKS encrypted matching was implemented using TenSEAL with $\text{poly_modulus_degree} = 8192$, $\text{coeff_mod_bit_sizes} = [60, 40, 40, 60]$, and $\text{global_scale} = 2^{40}$. The protected CKKS stage evaluated whether encrypted inner-product computation can preserve protected-domain verification behaviour.

The baseline results showed that CKKS encrypted matching preserved plain-domain performance. In unimodal verification, iris achieved the strongest baseline result with 1.80% EER and 0.9983 ROC-AUC. Face achieved 5.31% EER and 0.9865 ROC-AUC, while fingerprint achieved 16.30% EER and 0.9119 ROC-AUC. After cancellable transformation, protected-template performance became modality-dependent. The best protected iris configuration achieved 2.04% EER and 0.9979 ROC-AUC, the best protected face configuration achieved 5.74% EER and 0.9849 ROC-AUC, and the best protected fingerprint configuration achieved 18.43% EER and 0.8950 ROC-AUC.

Protected multimodal fusion significantly improved performance over protected unimodal systems. The protected Face + Iris fusion achieved 0.41% EER, Fingerprint + Iris achieved 1.41% EER, and Face + Fingerprint achieved 3.48% EER. The protected trimodal Face + Fingerprint + Iris system achieved the strongest result with 0.31% EER and 0.9999 ROC-AUC. Compared with the best protected unimodal system, the protected trimodal system provided an absolute EER gain of 1.73% and a relative recovery gain of approximately 84.80%. This

recovery occurs because different modalities degrade differently after cancellable transformation, so their errors are partially non-overlapping and score-level fusion combines complementary protected evidence.

Protected CKKS matching produced the same verification behaviour as protected plain matching for the evaluated fusion systems. This shows that the recognition degradation is mainly caused by cancellable transformation, while CKKS primarily adds computation-level privacy with runtime overhead rather than additional accuracy loss. Overall, the results establish that secure multimodal biometric verification should be evaluated as a layered system. Cancellable transformation supports template renewability and practical unlinkability, multimodal fusion recovers utility after protection, and CKKS encrypted matching preserves protected-domain similarity computation. The proposed framework therefore provides a structured basis for analysing recognition utility, template protection, unlinkability, encrypted computation, recovery gain, and runtime cost together.

Keywords: Biometric verification, multimodal biometrics, cancellable biometrics, biometric template protection, CKKS, homomorphic encryption, score-level fusion, unlinkability, privacy-preserving biometrics.

CHAPTER 1

Introduction

1.1 Background

Biometric authentication has become an important technology for identity verification because it uses physiological or behavioural traits such as face, fingerprint, iris, voice, gait, or signature. Unlike passwords, PINs, smart cards, or identity tokens, biometric traits are naturally associated with an individual and are difficult to forget, share, or lose. Because of these advantages, biometric authentication is widely used in mobile security, access control, banking, attendance monitoring, border control, and digital identity systems [1].

Despite these advantages, biometric authentication introduces a serious privacy and security challenge. A password or PIN can be changed after compromise, but a biometric trait such as a face, fingerprint, or iris cannot be replaced in the same way. If a biometric template is leaked, stolen, or misused, the risk may remain for a long time. Such compromise may lead to unauthorized matching, identity misuse, cross-database tracking, and loss of user privacy [2], [3].

In most modern biometric systems, the stored template is not the original biometric image. Instead, the image is converted into a compact numerical representation, often called a feature vector or embedding. Deep learning-based biometric systems commonly use such embeddings because they contain discriminative identity information. However, even though these embeddings are different from raw images, they still carry sensitive identity-related information. Therefore, if biometric embeddings are stored, transmitted, or matched without proper protection, they may still be misused [2]–[5].

A secure biometric template protection scheme is generally expected to satisfy four major requirements: revocability, unlinkability, irreversibility, and performance preservation. Revocability means that a compromised template can be cancelled and replaced with a new

protected version. Unlinkability means that templates generated from the same biometric trait for different applications should not be easily linked. Irreversibility means that recovering the original biometric information from the protected template should be computationally difficult. Performance preservation means that the protected system should still verify genuine users with acceptable accuracy [3]–[6].

Biometric systems may be unimodal or multimodal. A unimodal system uses only one biometric trait, such as face, fingerprint, or iris. Such systems are simple and practical, but their performance may be affected by the limitations of the selected modality. Face recognition may be affected by pose, illumination, expression, occlusion, and aging. Fingerprint recognition may be affected by partial impressions, dry skin, scars, pressure variation, and sensor quality. Iris recognition is generally highly discriminative, but it depends on proper image acquisition, segmentation, and illumination [7], [8].

Multimodal biometric systems address some of these limitations by combining two or more biometric traits. The main advantage of multimodal verification is complementarity. If one modality becomes weak or noisy, another modality may still provide useful identity evidence. Therefore, multimodal systems can improve robustness, population coverage, and verification performance compared with unimodal systems [7], [8].

Fusion is a central component of multimodal biometric verification. Fusion may be performed at sensor level, feature level, score level, or decision level. In this project, score-level fusion is used because it is practical for combining heterogeneous biometric modalities such as face, fingerprint, and iris. Each modality can be processed independently, and the resulting matching scores can then be normalized and combined. Score normalization is important because different biometric matchers may produce scores in different ranges. Without normalization, one modality may dominate the final fused decision even when it is not the most reliable [9], [10].

Although multimodal biometric systems improve recognition performance, they also increase the amount of sensitive biometric information processed by the system. A system that combines face, fingerprint, and iris handles more identity-bearing information than a unimodal system. Therefore, multimodal biometric verification requires stronger privacy protection. A secure multimodal biometric system should not only improve accuracy but should also protect biometric templates during storage, renewal, comparison, and matching [2]–[6].

Cancellable biometrics provides one approach to biometric template protection. In cancellable biometrics, the original biometric representation is transformed into a protected and key-dependent representation. If the protected template is compromised, another protected version can be generated using a different key. This supports revocability and renewability. However, cancellable transformation may also disturb the original embedding space and reduce verification performance. Therefore, cancellable biometric systems must be evaluated through a security–utility perspective [11]–[13].

Homomorphic encryption provides another privacy-preserving mechanism. It allows computation to be performed on encrypted data. In biometric verification, this means that similarity computation can be carried out without directly exposing the biometric template during matching. The CKKS homomorphic encryption scheme supports approximate arithmetic over real-valued vectors, making it suitable for deep biometric embeddings [14]. TenSEAL provides practical support for encrypted tensor operations and enables implementation of CKKS-based encrypted vector computation [15].

The present Major Project-II brings these directions together in a single secure multimodal verification framework. It studies face, fingerprint, and iris verification using cancellable template transformation, protected score-level fusion, same-key and cross-key evaluation, and CKKS encrypted matching. However, the contribution of this work is not merely the integration of existing techniques. The main contribution is the system-level analysis of how

template protection, multimodal fusion, unlinkability evaluation, encrypted computation, and runtime overhead interact in a privacy-preserving biometric verification pipeline.

1.2 Research Motivation

The motivation for this project comes from the need to design biometric verification systems that are not only accurate but also secure, revocable, unlinkable, and privacy-preserving. Many biometric systems are evaluated mainly on recognition accuracy. However, accuracy alone is not sufficient when the stored representation is a permanent and sensitive biometric template [2], [3].

If a biometric template is compromised, the user cannot simply replace the underlying biometric trait. This creates a long-term security risk. The problem becomes more serious in multimodal systems because multiple biometric traits are processed together. While multimodal fusion improves verification reliability, it also increases the privacy impact of template exposure [2]–[8].

Existing research has addressed different parts of this problem. Multimodal biometric systems have been studied for improving recognition performance and robustness [7], [8]. Score-level fusion has been studied for combining heterogeneous biometric matchers [9], [10]. Cancellable biometrics has been studied for revocable and application-specific template protection [11]–[13]. Homomorphic encryption has been studied for secure computation over encrypted biometric data [14], [15].

However, these directions are often evaluated separately. A multimodal biometric system may show high accuracy but may not address template renewability or unlinkability. A cancellable biometric system may protect templates but may reduce matching performance. A homomorphic encryption-based system may protect computation but may not provide revocation if a stored template is compromised [3]–[6], [11]–[15].

Therefore, there is a need to evaluate these mechanisms together under one common biometric verification protocol. This project is motivated by that gap. It investigates whether cancellable transformation, multimodal score-level fusion, same-key and cross-key unlinkability evaluation, and CKKS encrypted matching can jointly support secure and usable biometric verification.

1.3 Research Gap

The literature shows that biometric template protection, multimodal fusion, cancellable biometrics, and homomorphic encryption are important research areas. However, there remains a need for a unified system-level evaluation that studies these components together.

The main research gaps addressed in this work are as follows:

1. Many multimodal biometric studies focus mainly on improving recognition performance, but they do not deeply evaluate template revocability, renewability, and unlinkability [7]–[10].
2. Many cancellable biometric studies evaluate protected templates in unimodal settings, but they do not sufficiently analyse whether multimodal fusion can recover the utility loss caused by template protection [11]–[13].
3. Homomorphic encryption-based biometric systems protect the computation stage, but encryption alone does not provide template revocation or cross-application unlinkability [14], [15].
4. There is limited system-level analysis that jointly considers protected unimodal behaviour, protected multimodal fusion, same-key utility, cross-key unlinkability, CKKS encrypted matching, recovery gain, and runtime overhead [3]–[15].

The present work addresses this gap by evaluating cancellable transformation, protected score-level fusion, and CKKS encrypted matching under a common face–fingerprint–iris verification

framework. This makes it possible to analyse the system not only in terms of accuracy but also in terms of renewability, unlinkability, encrypted computation, recovery gain, and latency.

1.4 Problem Statement

Biometric templates are sensitive, permanent, and difficult to revoke after compromise. If a stored biometric representation is exposed, the user cannot easily replace the underlying biometric trait. This creates long-term privacy and security risks, especially when the same biometric trait is used across multiple applications or databases [2]–[6].

Multimodal biometric systems improve verification performance by combining multiple biometric traits, but they also process more sensitive identity-related information. Therefore, secure multimodal biometric verification requires mechanisms that can protect biometric templates, support renewability, reduce cross-key linkability, and enable privacy-preserving matching [7]–[10].

Major Project-I studied CKKS-based encrypted matching for multimodal biometric verification and showed that encrypted similarity computation can preserve plain-domain verification behaviour. However, encrypted matching mainly protects the computation stage. It does not directly provide template revocability, renewability, or unlinkability, which are central requirements of biometric template protection [3]–[6], [14], [15].

The problem addressed in this project is to extend the earlier CKKS-based multimodal biometric verification framework by incorporating cancellable biometric template transformation, same-key and cross-key evaluation, protected score-level fusion, and protected CKKS encrypted matching. The aim is to analyse whether a multimodal biometric system can maintain strong verification performance while also supporting template renewability, practical unlinkability, and computation-level privacy [11]–[15].

1.5 Research Questions

This project is guided by the following research questions:

1. How does cancellable template transformation affect the verification performance of face, fingerprint, and iris modalities?
2. Do protected templates retain sufficient verification utility when compared under the same transformation key?
3. Do templates generated using different transformation keys become difficult to link, thereby supporting practical unlinkability and renewability?
4. Can multimodal score-level fusion recover the utility loss introduced by cancellable template transformation?
5. Does CKKS encrypted matching preserve protected-domain verification behaviour after cancellable transformation?
6. What security–utility and latency trade-off is observed when cancellable biometrics, multimodal fusion, and CKKS encrypted matching are evaluated together?

These research questions position the project as an analytical system-level study rather than a simple implementation of existing methods.

1.6 Objectives of the Project

The main objectives of this project are as follows:

1. To extend the Major Project-I CKKS-based multimodal biometric verification framework by adding cancellable template protection for face, fingerprint, and iris modalities.
2. To evaluate binary and non-binary cancellable template transformations across different projection dimensions.
3. To analyse same-key protected verification performance as a measure of retained utility.
4. To analyse cross-key protected verification behaviour as empirical evidence of practical unlinkability and renewability.
5. To study modality-wise security–utility behaviour after template protection.

6. To evaluate whether protected multimodal score-level fusion can recover the performance degradation introduced by cancellable transformation.
7. To examine whether CKKS encrypted matching can preserve protected-domain verification behaviour.
8. To analyse the overall trade-off between verification accuracy, template protection, unlinkability, encrypted computation, and runtime overhead.

1.7 Research Contribution of the Project

The contribution of this work is not the development of a new biometric recognition model, a new cancellable transformation algorithm, or a new cryptographic scheme. Instead, the contribution lies in establishing a system-level security–utility framework for privacy-preserving multimodal biometric verification.

The major contributions of this project are as follows.

1. System-level secure multimodal verification framework:

This work presents a unified framework for evaluating privacy-preserving biometric verification using face, fingerprint, and iris modalities. The framework jointly considers template protection, multimodal fusion, unlinkability, encrypted matching, and runtime overhead. This is important because secure biometric verification cannot be judged only by recognition accuracy [2]–[10].

2. Cancellable template protection with modality-wise analysis:

The project applies key-dependent cancellable transformation to deep biometric embeddings and evaluates its effect on each modality. The analysis shows that face, fingerprint, and iris do not respond to template protection in the same way. Therefore, protected-template configuration should be modality-aware rather than uniform for all biometric traits [11]–[13].

3. Formal same-key and cross-key evaluation structure:

Same-key evaluation is used to measure whether protected templates retain verification utility. Cross-key evaluation is used to study whether templates generated from different transformation keys become difficult to link. This provides an empirical framework for analysing renewability and practical unlinkability [3]–[6], [11]–[13].

4. Fusion as a utility-recovery mechanism:

The work establishes that multimodal fusion after template protection is not only an accuracy-enhancement step. It acts as a utility-recovery mechanism because different modalities preserve different amounts of discriminative information after cancellable transformation. Therefore, the fused system can compensate for the weakness of individual protected matchers [7]–[13].

5. Protected CKKS encrypted matching:

The project evaluates CKKS encrypted matching after cancellable transformation. This shows whether encrypted-domain computation can preserve protected-domain verification behaviour. The analysis helps separate recognition loss from computational overhead [14], [15].

6. Integrated security–utility and latency interpretation:

The project analyses recognition performance, cross-key unlinkability, recovery gain, encrypted computation, and runtime overhead together. This provides a practical framework for selecting secure biometric configurations according to the required balance between accuracy, privacy, and computational cost [3]–[15].

The central system-level insight established by this work is that cancellable transformation, multimodal fusion, and CKKS encrypted matching should not be evaluated independently. Cancellable transformation provides renewability and unlinkability, fusion recovers part of the utility loss introduced by protection, and CKKS enables computation-level privacy without materially changing the protected-domain verification behaviour.

1.8 System-Level Insight Established by the Work

The main research insight of this project is that secure multimodal biometric verification should be evaluated as a layered system rather than as separate template protection, fusion, and encryption modules.

Cancellable transformation provides template-level protection by making biometric templates revocable and key-dependent. However, this protection may reduce unimodal verification performance because the original embedding space is modified [11]–[13]. Multimodal fusion helps recover part of this lost utility by combining complementary protected evidence from face, fingerprint, and iris [7]–[10]. CKKS encrypted matching then adds computation-level privacy by allowing similarity computation over encrypted protected templates [14], [15].

Thus, the final behaviour of the system depends on the interaction between five factors: protection strength, modality-specific degradation, fusion-based recovery, cross-key unlinkability, and encrypted matching overhead. This system-level interpretation is the main research value of the work. It shows that secure biometric verification should not be evaluated using accuracy alone, but through a combined analysis of utility, renewability, unlinkability, encrypted computation, and runtime cost [3]–[15].

1.9 Scope of the Project

This project focuses on biometric verification, where the system checks whether two biometric samples belong to the same claimed identity. It does not address large-scale biometric identification.

The work considers three biometric modalities: face, fingerprint, and iris. The scope includes deep embedding extraction, cancellable template transformation, binary and non-binary protected templates, projection dimension analysis, same-key and cross-key verification, score-level fusion, CKKS encrypted matching, recovery gain analysis, and runtime evaluation.

The system is evaluated using Equal Error Rate, ROC-AUC, accuracy, True Acceptance Rate at fixed False Acceptance Rate, recovery gain, and runtime. These metrics are used to analyse both recognition performance and privacy-preserving behaviour.

The project does not claim to provide a formal mathematical proof of irreversibility. It also does not address secure key management, side-channel attacks, adversarial attacks, malicious server behaviour, or fully encrypted end-to-end deployment. These limitations are outside the scope of the present work and are considered future research directions.

1.10 Organization of the Report

The remaining chapters of the report are organized as follows.

Chapter 2 presents the background and literature review related to biometric template security, multimodal biometric verification, score-level fusion, cancellable biometrics, same-key and cross-key unlinkability evaluation, and homomorphic encryption-based biometric matching. It also identifies the research gap addressed by the present work.

Chapter 3 presents the Major Project-I baseline framework. It describes dataset preparation, subject-level splitting, balanced verification-pair generation, deep embedding extraction, plain-domain matching, CKKS encrypted matching, and score-level fusion.

Chapter 4 presents the methodology of Major Project-II. It describes the proposed privacy-preserving framework, threat model, cancellable template transformation, binary and non-binary protected templates, projection dimension analysis, same-key and cross-key verification, protected fusion, and protected CKKS encrypted matching.

Chapter 5 describes the experimental setup and evaluation protocol. It includes dataset organization, subject-level splitting, verification-pair generation, experimental configurations, protection settings, CKKS parameters, evaluation metrics, and reporting strategy.

Chapter 6 presents the results and analysis. It discusses plain and CKKS baseline results, protected unimodal verification, same-key and cross-key behaviour, protected multimodal fusion, recovery gain, latency-over-security analysis, and security–utility trade-off.

Chapter 7 concludes the report. It summarizes the major findings, explains the research contribution, discusses limitations, and suggests future work.

CHAPTER 2

Literature Review

2.1 Introduction

Biometric verification has become an important research area because biometric traits provide a natural and convenient basis for identity authentication. Unlike passwords, PINs, or tokens, biometric traits are physically or behaviourally linked to an individual and are difficult to forget, share, or lose [1]. However, this permanence also creates a serious privacy and security concern. If a biometric template is compromised, the underlying biometric trait cannot be replaced in the same way as a password. Therefore, biometric templates must be protected against leakage, misuse, unauthorized matching, and cross-application linkage [2], [3].

The present work lies at the intersection of biometric template protection, multimodal biometric verification, score-level fusion, cancellable biometrics, unlinkability evaluation, and homomorphic encryption. Each of these areas addresses a different aspect of secure biometric system design. Template protection focuses on securing stored biometric representations. Multimodal biometrics improves verification reliability by combining multiple traits. Score-level fusion provides a practical way to combine heterogeneous biometric matchers. Cancellable biometrics supports revocable and key-dependent protected templates. Homomorphic encryption enables privacy-preserving computation over encrypted biometric data [3]–[15].

Although these areas have been studied individually, a practical privacy-preserving biometric system requires them to be evaluated together. A system that achieves high recognition accuracy may still be insecure if its templates are non-revocable or linkable across applications. Similarly, encrypted matching can protect the computation stage, but it does not automatically provide template renewability or unlinkability. Therefore, this chapter reviews the literature and identifies the need for a system-level security–utility framework that jointly evaluates

recognition performance, template protection, renewability, unlinkability, protected fusion, encrypted computation, and runtime overhead.

2.2 Biometric Authentication and Template Security

Biometric authentication verifies an individual using physiological or behavioural traits such as face, fingerprint, iris, voice, gait, or signature. Biometric traits are useful because they are naturally associated with the user and are difficult to forget or share [1]. However, the same permanence that makes biometrics useful also makes biometric compromise more serious than password compromise. A password can be changed, but a biometric trait cannot be replaced easily after exposure.

Ratha, Connell, and Bolle discussed several security and privacy risks in biometric systems, including template compromise, replay, substitution, and unauthorized use [2]. Jain, Nandakumar, and Nagar further emphasized that biometric templates contain identity-related information and must be protected because leakage may create long-term privacy risks [3].

A biometric template protection scheme is generally expected to satisfy four major requirements: revocability, unlinkability, irreversibility, and performance preservation. Revocability means that a compromised template can be cancelled and replaced with a new protected version. Unlinkability means that protected templates generated from the same biometric trait in different applications should not be easily associated with each other. Irreversibility means that recovering the original biometric information from the protected representation should be computationally difficult. Performance preservation means that the protected system should still verify genuine users with acceptable accuracy [3]–[6].

Rathgeb and Uhl reviewed biometric cryptosystems and cancellable biometrics as major directions for biometric template protection [4]. ISO/IEC 24745 identifies irreversibility, unlinkability, and renewability as important requirements for biometric information protection

[5]. Gomez-Barrero et al. proposed a general framework for evaluating unlinkability in biometric template protection systems [6].

These studies show that biometric systems should not be evaluated only through recognition accuracy. A secure biometric verification system must also be analysed in terms of template renewability, unlinkability, irreversibility, and utility preservation. This motivates the use of cancellable template transformation and same-key/cross-key evaluation in the present work.

2.3 Multimodal Biometric Verification

A unimodal biometric system uses only one biometric trait, such as face, fingerprint, or iris. Although unimodal systems are widely used, their performance may be affected by the limitations of the selected modality. Face recognition may be affected by pose, illumination, expression, occlusion, and aging [16]. Fingerprint recognition may be affected by dry skin, scars, pressure variation, partial impressions, and sensor quality [17]. Iris recognition is generally highly discriminative, but it depends on proper image acquisition, segmentation, and illumination [18].

Multimodal biometric systems address some of these limitations by combining two or more biometric traits. Ross and Jain showed that information fusion can improve biometric system reliability by combining evidence from multiple sources [7]. Ross, Nandakumar, and Jain further explained that multibiometric systems can improve robustness, population coverage, and recognition accuracy by reducing dependence on a single biometric source [8].

The main strength of multimodal verification is complementarity. If one biometric modality produces a weak or noisy matching score, another modality may still provide useful identity evidence. This makes multimodal systems more reliable than unimodal systems in practical verification scenarios [7], [8].

However, multimodal systems also increase privacy risk because they process more identity-bearing information. A system that uses face, fingerprint, and iris together stores or compares

multiple sensitive biometric representations. Therefore, multimodal biometric verification requires stronger privacy-preserving mechanisms than unimodal verification. This motivates the present work to combine multimodal verification with cancellable template protection and encrypted matching.

2.4 Score-Level Fusion in Multimodal Biometrics

Fusion is a central component of multimodal biometric verification. Fusion may be performed at sensor level, feature level, score level, or decision level. Sensor-level and feature-level fusion can be difficult when the biometric modalities are heterogeneous. Decision-level fusion is simple, but it may discard useful score information. Score-level fusion provides a practical balance because each modality can be processed independently and the resulting matching scores can be combined for the final decision [7]–[10].

Jain, Nandakumar, and Ross showed that score normalization is important in multimodal biometric systems because different matchers may produce scores in different ranges [9]. Without normalization, one modality may dominate the fused result even when it is not the most reliable. Nandakumar et al. further showed that score-level fusion can improve biometric decision-making when matcher outputs are combined appropriately [10].

In the present work, score-level fusion has a specific research role. It is not used only to improve plain-domain recognition performance. It is also evaluated after cancellable template transformation. This is important because template protection may reduce the discriminative strength of individual modalities. If the protected modalities retain complementary identity evidence, fusion can recover part of the utility loss caused by transformation.

Thus, score-level fusion is used in this work as a utility-recovery mechanism after template protection. This connects multimodal fusion directly with the security–utility analysis of the proposed framework.

2.5 Biometric Template Protection Approaches

Biometric template protection methods can be broadly grouped into biometric cryptosystems and feature-transformation-based methods [3], [4]. Biometric cryptosystems bind or derive cryptographic information from biometric features. Feature-transformation methods transform the original biometric representation into a protected representation before storage or matching. Cancellable biometrics belongs to the feature-transformation category. In cancellable biometrics, the original biometric representation is transformed using a key-dependent or application-specific function. If the protected template is compromised, a new protected version can be generated using another key. This makes cancellable biometrics suitable for revocability, renewability, and application-specific template protection [11]–[13].

Ratha et al. introduced cancellable fingerprint templates and showed that transformed biometric representations can be used for protected authentication [11]. Jin, Ling, and Goh proposed BioHashing, where biometric features are combined with tokenized random information to generate protected codes [12]. Patel, Ratha, and Chellappa reviewed cancellable biometrics and identified diversity, revocability, non-invertibility, and performance preservation as important requirements [13].

However, cancellable transformation naturally creates a security–utility trade-off. A stronger transformation may improve privacy and unlinkability, but it may also distort identity information and reduce verification performance. A weaker transformation may preserve accuracy but may provide limited protection. Therefore, cancellable biometrics must be evaluated by considering both recognition performance and security-oriented behaviour.

The present work follows this direction by applying random projection-based cancellable transformation to deep biometric embeddings and evaluating its effect through same-key utility, cross-key unlinkability, protected fusion performance, and runtime overhead.

2.6 Cancellable Biometrics for Deep Templates

Earlier cancellable biometric systems were often designed for handcrafted biometric features. However, modern biometric systems commonly use deep embeddings extracted from neural network models. These embeddings are compact and discriminative, but they still contain sensitive identity-related information. Therefore, protecting deep biometric templates is an important research direction.

Shahreza et al. benchmarked cancellable biometric methods for deep templates and emphasized the need to evaluate recognition performance, unlinkability, and irreversibility together [19]. Yang et al. reviewed feature extraction and learning approaches for cancellable biometrics and showed that cancellable biometric research is increasingly moving toward learned and deep representations [20].

These studies support the need to evaluate cancellable transformation on deep embeddings rather than only on handcrafted features. They also show that protected-template evaluation should be multidimensional. Accuracy alone is not sufficient; unlinkability, renewability, and the effect of transformation parameters must also be studied.

In the present work, 512-dimensional embeddings are used for face, fingerprint, and iris. Cancellable transformation is evaluated using different projection dimensions and both binary and non-binary protected representations. This allows the work to analyse how transformation strength affects different biometric modalities and how the resulting protected templates behave under same-key and cross-key conditions.

2.7 Same-Key and Cross-Key Evaluation

Same-key and cross-key evaluation are essential for analysing cancellable biometric systems. In same-key evaluation, both protected templates in a comparison pair are generated using the same transformation key. If the samples belong to the same subject, the protected templates should remain similar. If they belong to different subjects, they should remain dissimilar.

Therefore, same-key evaluation measures the verification utility of the protected representation.

In cross-key evaluation, the two protected templates are generated using different transformation keys. In this case, even templates from the same subject should not remain strongly matchable, because the protected representations are intended to be key-specific and renewable. If cross-key genuine and impostor score distributions overlap strongly, it indicates that templates generated under different keys are difficult to link. This provides empirical evidence of practical unlinkability [5], [6].

This distinction is important for a formal security–utility interpretation. High same-key performance alone does not prove template security, because protected templates may still be linkable across applications. Similarly, strong cross-key unlinkability is not useful if same-key verification performance becomes too weak for practical use. Therefore, both same-key and cross-key results must be interpreted together.

In the present work, same-key evaluation is used as a measure of retained utility, while cross-key evaluation is used as empirical evidence of renewability and practical unlinkability. This structure strengthens the security analysis of the proposed framework and avoids treating unlinkability as a purely descriptive result.

2.8 Homomorphic Encryption for Biometric Matching

Homomorphic encryption allows computation to be performed on encrypted data. After decryption, the result corresponds to the computation that would have been performed on the original data. This property is useful for biometric verification because similarity computation can be performed without directly exposing the biometric template during matching.

The CKKS scheme supports approximate arithmetic over encrypted real-valued numbers [14]. This makes it suitable for biometric embeddings, which are generally continuous-valued

vectors. TenSEAL provides practical support for encrypted tensor operations and enables CKKS-based encrypted vector computation [15].

In biometric matching, CKKS can be used to compute encrypted inner products or similarity scores. If encrypted computation preserves the same score behaviour as plain computation, then CKKS can provide computation-level privacy without reducing recognition accuracy.

However, homomorphic encryption alone does not satisfy all biometric template protection requirements. It protects the computation stage, but it does not automatically provide template revocation, renewability, or unlinkability. If the same biometric representation is reused across systems, encryption does not by itself prevent cross-application linkage or enable replacement of a compromised template.

For this reason, the present work applies CKKS encrypted matching after cancellable template transformation. In this layered design, cancellable biometrics provides template-level renewability and unlinkability, while CKKS provides computation-level privacy during similarity matching.

2.9 Privacy-Preserving Multimodal Biometric Systems

Recent research has explored privacy-preserving biometric systems that combine template protection and secure computation. Yang et al. reviewed homomorphic encryption for privacy-preserving biometrics and discussed its role in secure matching [21]. Gomez-Barrero et al. proposed multi-biometric template protection using homomorphic encryption, showing the relevance of encrypted computation in multibiometric systems [22]. Vallabhadas and Sandhya studied multimodal biometric template protection using local random projection and homomorphic encryption, which is closely related to projection-based protection with encrypted matching [23]. Rachapalli et al. studied multimodal cancellable biometric template protection and transformed-domain person verification [24].

These studies show that privacy-preserving multimodal biometrics is an active research direction. However, many existing works focus on only one or two aspects of the problem. Some focus on multimodal fusion, some on cancellable templates, and others on encrypted matching. Fewer studies evaluate how template protection, protected fusion, same-key/cross-key unlinkability, encrypted computation, and runtime overhead interact within a single verification framework.

Sharma et al. studied fully homomorphic encryption operators for score and decision fusion in biometric identification [25]. This supports the broader direction of encrypted biometric fusion, although the present work focuses on verification and CKKS-based protected similarity computation.

The present work is positioned within this research direction but contributes a system-level evaluation. It does not claim to introduce a new encryption scheme or a new cancellable transformation algorithm. Instead, it analyses how cancellable transformation, multimodal fusion, unlinkability evaluation, CKKS matching, and latency behave together in one face–fingerprint–iris verification framework.

2.10 Position of the Present Work

The present work is positioned as a system-level security–utility evaluation framework for privacy-preserving multimodal biometric verification. It differs from conventional multimodal biometric studies because it does not study fusion only as a method for improving recognition accuracy. It studies fusion after template protection and analyses whether fusion can recover the performance loss caused by cancellable transformation.

It differs from purely cancellable biometric studies because it does not evaluate protected templates only in a unimodal setting. Instead, protected face, fingerprint, and iris templates are evaluated individually and jointly. Same-key verification is used to measure retained utility, while cross-key verification is used to study practical unlinkability and renewability.

It also differs from purely homomorphic encryption-based biometric studies because encrypted matching is not treated as the only privacy-preserving mechanism. CKKS is used as a computation-level privacy layer after cancellable transformation. Therefore, the framework combines template-level protection and encrypted-domain similarity computation.

The work addresses the following system-level questions:

1. How does cancellable transformation affect face, fingerprint, and iris verification differently?
2. Do protected templates retain sufficient same-key verification utility?
3. Do cross-key protected templates become difficult to link?
4. Can multimodal fusion recover the utility loss introduced by template protection?
5. Does CKKS encrypted matching preserve protected-domain verification behaviour?
6. What trade-off exists between recognition performance, unlinkability, encrypted computation, and runtime overhead?

By addressing these questions, the present work provides an analytical framework for secure multimodal biometric verification rather than a simple integration of existing techniques.

2.11 Comparison of Related Studies with the Present Work

Table 2.1 Comparison of Related Studies with the Present Work

Study	Main Focus	Protection Method	Encrypted Matching	Unlinkability Analysis	Relevance to Present Work
Ratha et al. [2]	Biometric security and privacy	Security analysis	No	Conceptual	Establishes the need to protect biometric templates from compromise.
Jain et al. [3]	Template security	Template protection review	No	Conceptual	Supports revocability, unlinkability, irreversibility, and utility requirements.
Rathgeb and Uhl [4]	Biometric cryptosystems and	Template protection survey	No	Conceptual	Provides background on template

	cancellable biometrics				protection approaches.
ISO/IEC 24745 [5]	Biometric information protection	Standardized protection requirements	No	Yes	Defines renewability, unlinkability, and irreversibility.
Gomez-Barrero et al. [6]	Unlinkability evaluation	Template protection evaluation	No	Yes	Supports same-key and cross-key unlinkability analysis.
Ross and Jain [7]	Multimodal biometrics	Information fusion	No	No	Supports the use of multiple biometric traits.
Ross et al. [8]	Multibiometric systems	Multimodal framework	No	No	Supports robustness and complementarity of multimodal systems.
Jain et al. [9]	Score normalization	Score-level fusion	No	No	Supports normalization before score fusion.
Nandakumar et al. [10]	Score-level fusion	Fusion-based decision-making	No	No	Supports practical multimodal score fusion.
Ratha et al. [11]	Cancellable fingerprint templates	Cancellable transformation	No	Revocability discussed	Provides foundation for cancellable biometric templates.
Jin et al. [12]	BioHashing	Tokenized random transformation	No	Limited	Supports key-based protected representation.
Patel et al. [13]	Cancellable biometrics	Review of cancellable methods	No	Conceptual	Defines cancellable biometric requirements.
Cheon et al. [14]	Homomorphic encryption	CKKS scheme	Yes	No	Provides basis for encrypted real-valued computation.
TenSEAL [15]	Encrypted tensor operations	CKKS implementation	Yes	No	Supports practical encrypted vector computation.
Modality-specific biometric studies [16]–[18]	Face, fingerprint, and iris recognition	Recognition models and modality analysis	No	No	Support the selection of face, fingerprint, and iris as complementary modalities.

Shahreza et al. [19]	Deep cancellable templates	Benchmarking	No	Yes	Supports deep-template security–utility evaluation.
Yang et al. [20]	Cancellable biometrics	Feature extraction and learning review	No	Discussed	Supports cancellable protection for learned biometric representations.
Yang et al. [21]	HE for biometrics	Review	Yes	Discussed	Supports HE-based biometric privacy.
Gomez-Barrero et al. [22]	Multi-biometric template protection	Homomorphic encryption	Yes	Limited	Supports encrypted computation in multibiometric systems.
Vallabhadas and Sandhya [23]	Multimodal template security	Local random projection and HE	Yes	Limited	Related to projection-based protection with encrypted matching.
Rachapalli et al. [24]	Multimodal cancellable biometrics	Transformed-domain protection	No	Yes	Related to multimodal cancellable verification.
Sharma et al. [25]	Encrypted biometric fusion	FHE-based fusion	Yes	No	Supports encrypted fusion as an emerging direction.
Present Work	Secure face–fingerprint–iris verification	Cancellable random projection with binary and non-binary variants	Yes, CKKS	Yes, same-key and cross-key	Jointly evaluates protected fusion, CKKS matching, unlinkability, recovery gain, and latency–security trade-off.

This comparison shows that the present work is not simply an implementation of existing techniques. Its contribution lies in analysing how these techniques behave together under one controlled multimodal verification protocol. This system-level evaluation is important because secure biometric verification must be assessed in terms of accuracy, renewability, unlinkability, encrypted computation, recovery gain, and practical overhead.

2.12 Research Gap

The reviewed literature establishes that biometric templates require protection because they are sensitive, permanent, and difficult to revoke after compromise [2]–[6]. It also shows that multimodal biometrics can improve verification performance by combining complementary biometric traits [7], [8]. Cancellable biometrics supports revocable and key-dependent protected templates [11]–[13], while homomorphic encryption enables secure computation over encrypted biometric data [14], [15].

However, these research areas are often evaluated separately. Many multimodal biometric systems focus mainly on accuracy and robustness but do not deeply analyse template revocability and unlinkability. Many cancellable biometric systems evaluate protected templates in unimodal settings but do not study whether multimodal fusion can recover the utility loss caused by protection. Homomorphic encryption-based systems protect computation but do not automatically provide template renewal or cross-application unlinkability.

This creates a specific research gap: there is a need for a unified system-level evaluation of cancellable transformation, protected score-level fusion, same-key/cross-key unlinkability, CKKS encrypted matching, recovery gain, and runtime overhead under one common multimodal verification protocol.

The present work addresses this gap by evaluating face, fingerprint, and iris under the same subject-level split and balanced verification-pair protocol. It studies how protected templates behave across modalities, how same-key and cross-key results reflect utility and unlinkability, how fusion compensates for transformation-induced degradation, and how CKKS encrypted matching affects the final protected system.

2.13 Literature-to-Design Mapping

To make the connection between the literature and the proposed methodology clear, Table 2.2 summarizes how each research direction motivates a design choice in the present work.

Table 2.2 Literature-to-Design Mapping

Literature Finding	Limitation Identified	Design Choice in Present Work
Biometric templates are sensitive and difficult to revoke [2], [3]	Plain templates create long-term privacy risk	Use cancellable template transformation
Template protection requires renewability and unlinkability [3]–[6]	Accuracy alone is not sufficient	Use same-key and cross-key evaluation
Multimodal systems improve robustness through complementary evidence [7], [8]	Multiple traits increase privacy exposure	Combine fusion with protection layers
Score-level fusion is practical for heterogeneous matchers [9], [10]	Fusion alone does not protect templates	Apply fusion after cancellable transformation
Cancellable biometrics supports revocable templates [11]–[13]	Transformation may reduce utility	Analyse security–utility trade-off
Deep templates remain identity-bearing [19], [20]	Deep embeddings require protection	Protect 512-D deep embeddings
CKKS supports encrypted real-valued computation [14], [15]	Encryption alone does not provide renewability	Apply CKKS after cancellable protection
Privacy-preserving multimodal systems need integrated evaluation [21]–[25]	Existing studies often evaluate components separately	Establish a system-level evaluation framework

This mapping shows that the methodology of the present project is not arbitrary. Each design choice is derived from a specific limitation identified in the literature.

2.14 Summary

This chapter reviewed the literature related to privacy-preserving multimodal biometric verification. It first discussed biometric template security and showed that biometric templates must be protected because they are sensitive, permanent, and difficult to revoke [2]–[6]. It then reviewed multimodal biometric verification and score-level fusion, which support the use of face, fingerprint, and iris with normalized score-level fusion [7]–[10].

The chapter also reviewed cancellable biometrics, same-key/cross-key evaluation, and CKKS-based encrypted matching. Cancellable biometrics supports revocability and renewability [11]–[13]. Same-key and cross-key evaluation provide a practical way to analyse retained utility and unlinkability [5], [6]. CKKS homomorphic encryption supports encrypted computation over real-valued biometric embeddings and is therefore suitable for privacy-preserving similarity matching [14], [15].

The literature shows that multimodal biometrics, cancellable biometrics, and homomorphic encryption have each been studied individually, but their combined evaluation under one common verification protocol remains an important gap. The present work addresses this gap by evaluating cancellable template protection, protected score-level fusion, same-key/cross-key unlinkability, protected CKKS matching, recovery gain, and latency-security trade-off in a single face–fingerprint–iris framework.

Most importantly, this chapter establishes the research position of the project. The work is not framed as a simple integration of existing methods. It is framed as a system-level security–utility evaluation that analyses how template protection, multimodal fusion, unlinkability, encrypted computation, and runtime overhead interact in secure multimodal biometric verification.

CHAPTER 3

Major Project-1 Baseline Framework

3.1 Introduction

Major Project-II is developed as an extension of the baseline multimodal biometric verification framework established in Major Project-I. Therefore, before introducing cancellable template transformation, same-key/cross-key evaluation, protected fusion, and protected CKKS matching, it is necessary to clearly define the baseline system on which the proposed security–utility framework is built.

Major Project-I studied multimodal biometric verification using face, fingerprint, and iris modalities. The purpose of that phase was to establish a common verification protocol and to evaluate whether CKKS encrypted matching could preserve plain-domain biometric matching behaviour. The baseline framework included multimodal dataset preparation, subject-level splitting, balanced verification-pair generation, deep embedding extraction, plain-domain matching, CKKS encrypted matching, score-level fusion, and biometric performance evaluation.

This baseline is important because Major Project-II does not propose an isolated or unrelated system. Instead, it extends the earlier framework by adding template-level protection and a more formal security analysis. The plain-domain baseline provides the unprotected reference. CKKS encrypted matching provides the computation-level privacy reference. Major Project-II then adds cancellable transformation to address template renewability and unlinkability, which are central requirements of biometric template protection [3]–[6], [11]–[13].

Thus, Major Project-I is not treated as a separate preliminary implementation. It provides the controlled experimental foundation required to evaluate the system-level behaviour of privacy-preserving multimodal biometric verification.

3.2 Role of Major Project-I in the Overall Research Framework

Major Project-I served as the methodological foundation for the complete work. It established the data organization, embedding pipeline, verification-pair protocol, matching strategy, encrypted computation procedure, and score-level fusion method later reused in Major Project-II.

The role of the baseline framework is threefold.

First, it provides a controlled multimodal verification protocol. The same subject-level split and balanced pair-generation strategy were used across face, fingerprint, and iris. This allows fair comparison between unimodal, bimodal, and trimodal systems. It also reduces the risk of data leakage because identities used for training, validation, and testing are separated at the subject level.

Second, it establishes plain-domain and CKKS encrypted-domain reference performance. CKKS supports approximate arithmetic over encrypted real-valued vectors and is therefore suitable for similarity computation on deep biometric embeddings [14]. TenSEAL provides practical support for CKKS-based encrypted tensor operations [15]. By comparing plain-domain and CKKS-domain scores, Major Project-I evaluates whether encrypted computation changes the verification behaviour.

Third, it creates the reference point for the security extension in Major Project-II. Once the plain and CKKS baseline results are known, the effect of cancellable transformation can be measured more clearly. Major Project-II can then analyse how protected templates behave under same-key and cross-key conditions, whether protected fusion recovers utility loss, and whether CKKS matching preserves protected-domain verification behaviour.

Therefore, the baseline is essential for the research contribution of the present project. It enables the later work to move beyond simple implementation and toward a system-level security–utility analysis.

3.3 Baseline Framework Overview

The baseline system was designed for biometric verification using three physiological modalities: face, fingerprint, and iris. The task considered in this work is verification, where the system checks whether two biometric samples belong to the same claimed identity. This differs from identification, where the system searches for a subject from a larger enrolled database.

The baseline framework consisted of the following stages:

1. collection of face, fingerprint, and iris samples from public biometric datasets;
2. construction of a virtually subject-aligned multimodal dataset;
3. subject-level train, validation, and test split;
4. generation of balanced genuine and impostor verification pairs;
5. extraction of deep embeddings for each modality;
6. plain-domain similarity matching using cosine similarity or normalized dot product;
7. CKKS encrypted-domain similarity computation;
8. score-level fusion for bimodal and trimodal verification;
9. evaluation using standard biometric metrics.

The use of a common verification protocol is important because multimodal systems should be compared under the same experimental conditions. Prior work on multimodal biometrics has shown that combining multiple biometric traits can improve system reliability and recognition performance [7], [8]. Score-level fusion is particularly suitable for heterogeneous modalities because each matcher can operate independently and only the scores need to be combined [9], [10].

The baseline framework therefore supports two purposes. It provides the recognition reference for face, fingerprint, and iris verification, and it provides the controlled structure needed for later protected-domain evaluation.

3.4 Multimodal Dataset Preparation

A virtually subject-aligned multimodal dataset was prepared using three public biometric datasets. A virtual alignment strategy was used because a single large-scale public dataset containing aligned face, fingerprint, and iris samples from the same individuals was not available for the required experimental setting.

Face samples were taken from VGGFace2, which provides large-scale face images with variation in pose, age, illumination, and imaging conditions [26]. Fingerprint samples were taken from the SOCOFing dataset, which provides fingerprint images for biometric research [27]. Iris samples were taken from CASIA-Iris-Thousand, which is widely used for iris recognition research [28].

The final dataset contained 600 virtual subjects. Each subject contained 10 samples per modality. Therefore, each subject had 10 face images, 10 fingerprint images, and 10 iris images.

Overall, the dataset contained 18,000 biometric samples.

$$600 \times 10 \times 3 = 18,000$$

Table 3.1 Dataset Composition Used in the Baseline Framework

Component	Description
Number of modalities	3
Modalities	Face, Fingerprint, Iris
Number of virtual subjects	600
Samples per modality per subject	10
Total samples per subject	30
Total biometric samples	18,000

The virtual subject-aligned structure enables controlled multimodal fusion experiments. Although the subjects are virtually aligned rather than naturally collected from the same individuals, the protocol allows consistent comparison of unimodal, bimodal, and trimodal systems. This is useful for analysing system-level behaviour under identical pair-generation and evaluation conditions.

3.5 Subject-Level Split and Verification Pair Protocol

A subject-level split was used to avoid identity leakage between training, validation, and testing. In biometric verification, subject-level separation is important because sample-level splitting may allow different samples of the same subject to appear across different splits, which can produce overly optimistic results.

The 600 virtual subjects were divided into training, validation, and testing sets. The training set was used for model preparation and embedding generation. The validation set was used for score normalization and fusion-weight selection. The test set was used only for final performance reporting.

Table 3.2 Subject-Level Data Split

Split	Number of Subjects	Purpose
Training	480	Model training and embedding preparation
Validation	60	Score normalization and fusion weight selection
Testing	60	Final performance evaluation

For validation and testing, balanced verification pairs were generated. A genuine pair contains two samples from the same subject, while an impostor pair contains samples from different subjects. Balanced pair generation was used so that the number of genuine and impostor comparisons remained equal.

Each validation and test split contained 5,400 pairs per modality, with 2,700 genuine and 2,700 impostor pairs.

Table 3.3 Balanced Verification Pair Protocol

Split	Genuine Pairs	Impostor Pairs	Total Pairs
Validation	2,700	2,700	5,400
Testing	2,700	2,700	5,400

The balanced pair protocol is important for the overall research design. It allows the same comparison pairs to be used across plain matching, CKKS matching, protected matching, and protected fusion. Therefore, performance changes observed in Major Project-II can be attributed to the protection and encryption mechanisms rather than changes in the evaluation protocol.

3.6 Deep Embedding Extraction

The baseline framework used deep biometric embeddings for all three modalities. Each biometric image was converted into a fixed-dimensional numerical representation before matching. Deep embeddings are widely used in modern biometric systems because they capture discriminative identity information more effectively than many handcrafted representations.

Different feature extractors were used for different biometric modalities because face, fingerprint, and iris images have different visual structures. Face embeddings were extracted using an InceptionResnetV1 model pretrained on VGGFace2. Fingerprint embeddings were extracted using a ResNet-based model. Iris embeddings were extracted using an EfficientNet-B3-based model.

Table 3.4 Embedding Extraction Models

Modality	Feature Extractor	Embedding Representation
Face	InceptionResnetV1 pretrained on VGGFace2	Deep face embedding
Fingerprint	ResNet-based fingerprint model	Deep fingerprint embedding
Iris	EfficientNet-B3-based iris model	Deep iris embedding

The extracted embeddings were normalized before similarity computation. Normalization is important because cosine similarity and normalized dot product assume comparable vector magnitudes. The normalized embeddings also provide a consistent input representation for later cancellable transformation.

This embedding stage is central to the complete framework. In Major Project-I, embeddings are used for plain and CKKS matching. In Major Project-II, the same embeddings are transformed into protected templates using key-dependent cancellable projection. Therefore, the embedding pipeline links the baseline recognition system with the later template protection system.

3.7 Plain-Domain Matching

Plain-domain matching was used as the unprotected reference system. In this setting, biometric embeddings were compared directly without cancellable transformation or encryption. The similarity between two normalized embeddings was computed using cosine similarity or normalized dot product.

For two normalized embeddings x_i and x_j , the similarity score is given by:

$$s(x_i, x_j) = x_i^T x_j$$

A higher score indicates that the two samples are more likely to belong to the same subject, while a lower score indicates an impostor comparison.

Plain-domain matching is necessary because it establishes the maximum baseline utility of the extracted embeddings before privacy-preserving transformations are applied. It does not provide template protection, but it provides the reference against which CKKS encrypted matching and later protected matching can be compared.

This distinction is important for the system-level analysis. If protected or encrypted matching shows performance changes, those changes must be interpreted relative to the plain-domain baseline.

3.8 CKKS Encrypted Matching

CKKS encrypted matching was evaluated as the computation-level privacy layer in Major Project-I. Homomorphic encryption allows similarity computation to be performed on encrypted data. The CKKS scheme is suitable for approximate arithmetic over real-valued vectors, which makes it appropriate for biometric embeddings [14]. TenSEAL was used to implement CKKS-based encrypted vector computation [15].

In the baseline framework, similarity computation was performed using encrypted vector operations. The objective was to verify whether encrypted inner-product computation preserves the same matching behaviour as plain-domain similarity computation.

The CKKS parameter configuration used in the baseline framework . CKKS encrypted matching protects the computation stage by allowing similarity scores to be computed without directly exposing the encrypted vector during matching. However, CKKS alone does not solve all biometric template protection requirements. It does not automatically provide template revocation, renewability, or cross-key unlinkability. These requirements are central to biometric template protection [3]–[6].

This limitation is one of the main reasons Major Project-II introduces cancellable transformation before encrypted matching. In the complete framework, cancellable biometrics provides template-level protection, while CKKS provides computation-level privacy.

3.9 Score-Level Fusion

Score-level fusion was used to combine the matching scores of multiple biometric modalities. This fusion level was selected because face, fingerprint, and iris are heterogeneous modalities and can be processed independently before their scores are combined. Score-level fusion is widely used in multimodal biometric systems because it offers a practical balance between information richness and implementation simplicity [7]–[10].

Before fusion, scores were normalized to make them comparable across modalities. Score normalization is necessary because different matchers may produce scores with different ranges and distributions [9]. After normalization, weighted score-level fusion was applied.

For a bimodal system, the fused score is given by:

$$S_f = w_1 S_1 + w_2 S_2$$

For a trimodal system, the fused score is given by:

$$S_f = w_1 S_1 + w_2 S_2 + w_3 S_3$$

where S_1 , S_2 , and S_3 represent modality-specific matching scores, and w_1 , w_2 , and w_3 represent fusion weights. The weights were selected using the validation set and then applied to the test set.

Table 3.5 Fusion Systems Evaluated in the Baseline Framework

Fusion Type	Modalities Used
Bimodal	Face + Fingerprint
Bimodal	Face + Iris
Bimodal	Fingerprint + Iris
Trimodal	Face + Fingerprint + Iris

In Major Project-I, fusion was used to evaluate the benefit of combining complementary biometric modalities. In Major Project-II, the same fusion strategy is extended to protected templates. This allows the work to examine whether fusion can recover performance loss introduced by cancellable transformation. Therefore, fusion is not only a baseline accuracy-improvement method; it becomes a key component of the later security–utility analysis.

3.10 Baseline Results of Major Project-I

The baseline results showed that the three modalities had different verification strengths. Iris achieved the strongest unimodal performance, followed by face, while fingerprint showed weaker performance compared with the other two modalities. This modality-specific behaviour is important because the later protected evaluation also shows that cancellable transformation affects each modality differently.

The CKKS encrypted results preserved the plain-domain verification behaviour. This indicates that encrypted similarity computation did not materially alter the matching scores. This result is expected because CKKS matching follows the same inner-product structure as the plain-domain matcher [14], [15].

The baseline fusion results further showed that combining modalities improved verification performance compared with unimodal systems. Fusion systems involving iris performed strongly because iris contributed highly discriminative evidence, while face and fingerprint added complementary identity information. This supports the use of score-level fusion as a meaningful component of the later protected framework [7]–[10].

3.11 Interpretation of Baseline Results

The baseline results provide three important observations for the complete research framework. First, the three modalities do not have equal baseline strength. Iris provides the strongest unimodal performance, face provides strong but comparatively weaker performance, and fingerprint is more challenging. This observation is important because protected-template performance is also expected to be modality-dependent. Therefore, the later security–utility analysis should not assume that one protection setting will behave equally across all modalities. Second, CKKS encrypted matching preserves plain-domain verification behaviour. This shows that encrypted computation can support biometric similarity matching without materially reducing recognition performance. However, this result should not be interpreted as complete biometric template protection. CKKS protects the computation stage, but it does not by itself provide revocability, renewability, or unlinkability [3]–[6], [14], [15].

Third, score-level fusion improves verification performance by combining complementary biometric evidence. This confirms the value of multimodal fusion. More importantly, it motivates the protected fusion analysis in Major Project-II. If different modalities retain different levels of discriminative information after cancellable transformation, fusion may recover part of the lost utility.

Thus, the baseline results establish the reference behaviour required for the system-level security–utility evaluation in Major Project-II.

3.12 Significance and Limitations of the Baseline Framework

The Major Project-I baseline is significant because it establishes a controlled face–fingerprint–iris verification protocol. It provides common data organization, subject-level splitting, balanced verification pairs, embedding extraction, plain-domain matching, CKKS encrypted matching, score-level fusion, and performance evaluation.

This baseline makes the later protected evaluation meaningful. Since the same subjects, pairs, embeddings, and fusion protocol are reused, performance differences in Major Project-II can

be linked to cancellable transformation, cross-key behaviour, protected fusion, and encrypted protected matching rather than to changes in the experimental setup.

However, the baseline also has clear limitations. Plain-domain matching does not protect stored biometric templates. CKKS encrypted matching protects the similarity computation stage, but it does not provide template revocation or unlinkability. If a biometric embedding or protected representation is reused across systems, encryption alone does not prevent cross-application linkage. Similarly, if a template is compromised, CKKS does not by itself allow the template to be cancelled and regenerated using a new key.

These limitations motivate the transition from Major Project-I to Major Project-II. The second phase introduces cancellable template transformation, same-key and cross-key evaluation, protected score-level fusion, protected CKKS matching, and security–utility trade-off analysis. This transforms the work from an encrypted matching baseline into a broader system-level framework for secure multimodal biometric verification.

3.13 Summary

This chapter presented the Major Project-I baseline framework. The baseline included multimodal dataset preparation, subject-level splitting, balanced pair generation, deep embedding extraction, plain-domain similarity matching, CKKS encrypted matching, score-level fusion, and biometric performance evaluation.

The baseline established three important foundations for the present work. First, it provided a common face–fingerprint–iris verification protocol. Second, it showed that CKKS encrypted matching can preserve plain-domain verification behaviour. Third, it showed that score-level fusion improves verification performance by combining complementary biometric evidence.

At the same time, the baseline did not fully address template revocability, renewability, or cross-key unlinkability. These limitations provide the motivation for Major Project-II. The next chapter therefore presents the proposed methodology, where cancellable template

transformation, same-key and cross-key evaluation, protected score-level fusion, and CKKS encrypted matching over protected templates are introduced as part of a system-level security–utility framework.

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CHAPTER 4

Methodology Of Major Project-II

4.1 Introduction

Major Project-II extends the baseline multimodal biometric verification framework developed in Major Project-I by introducing cancellable template transformation, same-key and cross-key evaluation, protected multimodal fusion, and CKKS encrypted matching over protected templates. The aim of this chapter is to present the proposed methodology as a system-level security–utility framework rather than as a sequence of independently added techniques.

Major Project-I established the baseline face–fingerprint–iris verification pipeline using plain-domain matching, CKKS encrypted matching, and score-level fusion. That phase showed that CKKS encrypted matching can preserve plain-domain verification behaviour. However, CKKS mainly protects the computation stage. It does not directly provide template revocability, renewability, or unlinkability, which are central requirements of biometric template protection [3]–[6].

To address this limitation, Major Project-II introduces cancellable biometric template transformation before matching. The transformed templates are evaluated under same-key and cross-key conditions. Same-key evaluation measures whether protected templates retain verification utility, while cross-key evaluation measures whether templates generated under different keys become difficult to link. Protected scores are then fused to study whether multimodal fusion can recover the utility loss introduced by template protection. Finally, CKKS encrypted matching is applied to protected templates to examine whether computation-level privacy can be added without changing protected-domain verification behaviour [14], [15].

Thus, the methodology is designed as a layered privacy-preserving verification framework. Cancellable transformation provides template-level protection, same-key/cross-key evaluation measures utility and unlinkability, multimodal fusion provides utility recovery, and CKKS encryption provides computation-level privacy. This structure supports the research contribution of the work as a system-level security–utility evaluation of secure multimodal biometric verification.

4.2 Overview of the Proposed Methodology

The proposed methodology evaluates privacy-preserving multimodal biometric verification using three modalities: face, fingerprint, and iris. The framework uses the same dataset organization, subject-level split, balanced pair protocol, and embedding pipeline established in the baseline framework. This ensures that plain, protected, fused, and encrypted results are evaluated under comparable conditions.

The complete methodology consists of the following stages:

1. Deep embeddings are obtained for face, fingerprint, and iris samples.
2. The embeddings are normalized to provide comparable vector representations.
3. Cancellable transformation is applied using key-dependent random projection.
4. Protected templates are generated in binary and non-binary forms.
5. Same-key verification is performed to measure protected-template utility.
6. Cross-key verification is performed to evaluate renewability and practical unlinkability.
7. Protected modality scores are normalized and fused using score-level fusion.
8. CKKS encrypted matching is applied to protected templates.
9. The complete system is evaluated using EER, ROC-AUC, accuracy, TAR@FAR, recovery gain, unlinkability behaviour, and runtime.

The methodological contribution of this work lies in evaluating these stages as an interdependent privacy-preserving system rather than as independent modules. Cancellable

transformation is evaluated not only by its effect on unimodal accuracy but also by its influence on fusion recovery and cross-key unlinkability. Similarly, CKKS encrypted matching is not treated as a standalone security layer; it is evaluated after template protection to determine whether computation-level privacy can be added without changing protected-domain verification behaviour.

The overall workflow of the proposed framework is illustrated in Fig. 4.1. As shown in the figure, the methodology progresses from multimodal dataset preparation and deep embedding extraction to cancellable template generation, same-key/cross-key evaluation, protected fusion, CKKS encrypted matching, and final security–utility analysis.

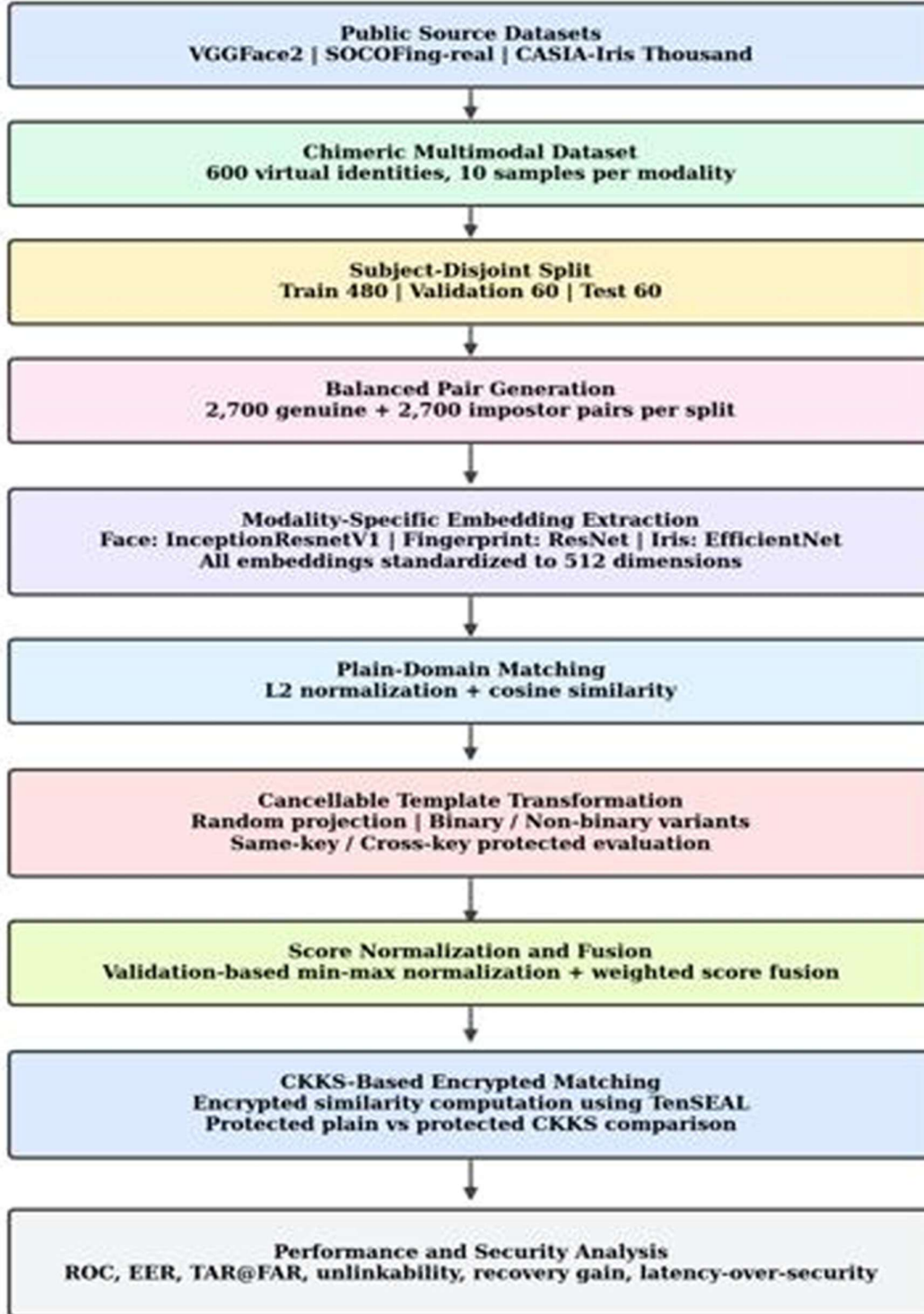


Fig. 4.1. Overall framework of the proposed privacy-preserving multimodal biometric verification system using cancellable template transformation and CKKS encrypted matching.

The methodology is designed to answer a central system-level question: can a multimodal biometric verification system maintain strong utility while also supporting template protection, renewability, unlinkability, encrypted matching, and acceptable computational cost?

4.3 Threat Model and Security Assumptions

The proposed framework assumes a privacy-sensitive biometric verification environment in which biometric templates may be stored, transformed, transmitted, or matched by a verification server. The protected assets in this framework are:

1. original biometric embeddings;
2. cancellable protected templates;
3. transformation keys;
4. similarity scores produced during matching;
5. modality-wise and fused verification decisions.

The threat model considers an **honest-but-curious matching server**. The server is assumed to follow the prescribed verification protocol, but it may attempt to infer identity-related information from stored templates, protected representations, matching scores, or repeated comparisons. The server may also attempt to determine whether two protected templates generated under different transformation keys belong to the same subject.

This threat model is relevant because biometric templates are sensitive and permanent. If a biometric representation is exposed, the user cannot replace the underlying biometric trait like a password. Therefore, template protection must support renewability and unlinkability [2]–[6].

The framework considers two main risks. First, a protected template may be compromised. In that case, the system should be able to revoke the old protected template and issue a new protected version using a different transformation key. This corresponds to the requirement of renewability [5], [13]. Second, protected templates generated from the same biometric trait

under different keys may be compared by an attacker. If such templates remain strongly matchable, they may allow cross-application linkage. Therefore, cross-key unlinkability must be evaluated [5], [6].

The threat model assumes that the transformation key is not intentionally disclosed to the attacker. It also assumes that the biometric feature extractors operate correctly and that the matching server does not maliciously alter the protocol. The present work does not claim protection against all possible attacks. It does not provide a formal mathematical proof of irreversibility, side-channel resistance analysis, adversarial attack evaluation, or a complete secure key-management protocol. These aspects are outside the scope of the present project and are treated as future work.

Within this threat model, the objective is to empirically evaluate whether cancellable transformation, protected fusion, and CKKS encrypted matching can jointly support verification utility, template renewability, practical unlinkability, and computation-level privacy.

Table 4.1 Threat Model Summary

Component	Description
System setting	Privacy-sensitive biometric verification
Server model	Honest-but-curious matching server
Protected assets	Original embeddings, protected templates, transformation keys, scores, decisions
Attacker goal	Infer identity, misuse templates, link templates across keys
Key assumption	Transformation keys are not disclosed
Utility evaluation	Same-key verification
Unlinkability evaluation	Cross-key verification
Computation privacy	CKKS encrypted matching
Out of scope	Formal irreversibility proof, malicious server, side-channel attack, secure key management

4.4 Input Embeddings and Normalization

The input to the proposed protection framework consists of deep biometric embeddings extracted for face, fingerprint, and iris. These embeddings are inherited from the Major Project-

I baseline framework. Each embedding represents identity-related information in numerical vector form.

Before applying cancellable transformation, embeddings are normalized. Normalization is important because biometric embeddings may have different magnitudes or distributions across modalities. Similarity matching using cosine similarity or normalized dot product assumes comparable vector magnitudes. Normalization also provides stable input for random projection-based transformation.

Let $x \in \mathbb{R}^d$ represent an original biometric embedding of dimension d . The normalized embedding is represented as:

$$\hat{x} = \frac{x}{\|x\|_2}$$

where $\|x\|_2$ is the L2 norm of the embedding vector.

The normalized embedding $\hat{x}$ is then used as the input to the cancellable transformation stage. This step ensures that the transformation operates on a consistent representation and that the resulting protected templates can be compared using similarity measures.

4.5 Cancellable Template Transformation

Cancellable biometric transformation is used to convert the original biometric embedding into a protected, key-dependent representation. The purpose of this transformation is to support revocability and renewability. If a protected template is compromised, a new protected template can be generated using a different transformation key [11]–[13].

In this work, cancellable transformation is implemented using random projection. For a normalized biometric embedding $\hat{x} \in \mathbb{R}^d$, a key-dependent random projection matrix $R_k \in \mathbb{R}^{d \times m}$ is generated, where k denotes the transformation key and m denotes the protected-template dimension. The protected non-binary template z_k is computed as:

$$z_k = \hat{x}^T R_k$$

where $z_k \in \mathbb{R}^m$.

The transformation is key-dependent because different keys generate different projection matrices. Therefore, the same biometric embedding can produce different protected templates under different keys. This supports renewability and diversity.

The transformation also changes the original embedding space. This is useful for template protection, but it may reduce verification performance if too much discriminative information is lost. Therefore, the project evaluates multiple projection dimensions and both binary and non-binary forms to analyse the security–utility trade-off.

4.6 Binary and Non-Binary Protected Templates

Two types of protected representations are evaluated: non-binary protected templates and binary protected templates.

In the non-binary case, the projected vector z_k is directly used as the protected representation:

$$T_k^{NB} = z_k$$

where T_k^{NB} denotes the non-binary protected template generated under key k .

In the binary case, the projected vector is converted into a binary code using a sign-based binarization function:

$$T_k^B = \text{sign}(z_k)$$

where T_k^B denotes the binary protected template.

The non-binary representation may preserve more similarity information because it retains continuous-valued projection outputs. However, it may also retain more structural information from the original embedding. The binary representation provides a more compact and discretized form, but binarization may remove useful matching information and reduce verification performance.

Therefore, both representations are evaluated. This allows the project to analyse whether binary or non-binary templates provide a better balance between protected utility, unlinkability, and runtime behaviour.

4.7 Projection Dimension Analysis

Projection dimension is an important parameter in cancellable template transformation. A lower projection dimension may provide stronger compression and may reduce linkability, but it may also remove useful identity information. A higher projection dimension may preserve more discriminative information, but it may provide weaker protection and higher computational cost.

In this work, the following protected-template dimensions are evaluated:

$$m \in \{64, 128, 256, 512, 1024\}$$

The purpose of evaluating multiple dimensions is to analyse the security–utility trade-off. If the projection dimension is too small, same-key verification performance may degrade. If it is too large, the protected representation may preserve more structure from the original embedding. Therefore, the optimal dimension must be selected by considering both verification utility and cross-key unlinkability.

The dimension analysis is performed separately for face, fingerprint, and iris because different modalities may respond differently to cancellable transformation. This modality-wise analysis is important because one fixed protection setting may not be suitable for all biometric traits.

4.8 Same-Key and Cross-Key Verification

Same-key and cross-key verification are used to formally evaluate the protected templates.

In **same-key verification**, both templates in a comparison pair are generated using the same transformation key. For two normalized embeddings $\hat{x}_i$ and $\hat{x}_j$, the same-key protected templates are:

$$T_k(\hat{x}_i), T_k(\hat{x}_j)$$

The protected similarity score is computed as:

$$s_{same} = sim(T_k(\hat{x}_i), T_k(\hat{x}_j))$$

Same-key evaluation measures protected-template utility. If the two samples belong to the same subject, the protected templates should remain similar. If the samples belong to different subjects, the protected templates should remain dissimilar. Therefore, strong same-key performance indicates that the cancellable transformation has preserved sufficient discriminative information for verification.

In **cross-key verification**, the two templates are generated using different transformation keys:

$$T_{k_1}(\hat{x}_i), T_{k_2}(\hat{x}_j), k_1 \neq k_2$$

The cross-key protected similarity score is computed as:

$$s_{cross} = sim(T_{k_1}(\hat{x}_i), T_{k_2}(\hat{x}_j))$$

Cross-key evaluation measures practical unlinkability and renewability. If templates generated under different keys remain strongly matchable, they may allow cross-application linkage. A desirable cancellable biometric system should show good same-key utility and weak cross-key matchability [5], [6].

The distinction between same-key and cross-key evaluation is illustrated in Fig. 4.2. As shown in the figure, same-key comparison measures retained verification utility under the same transformation condition, whereas cross-key comparison examines whether renewed templates generated under different keys become difficult to associate.

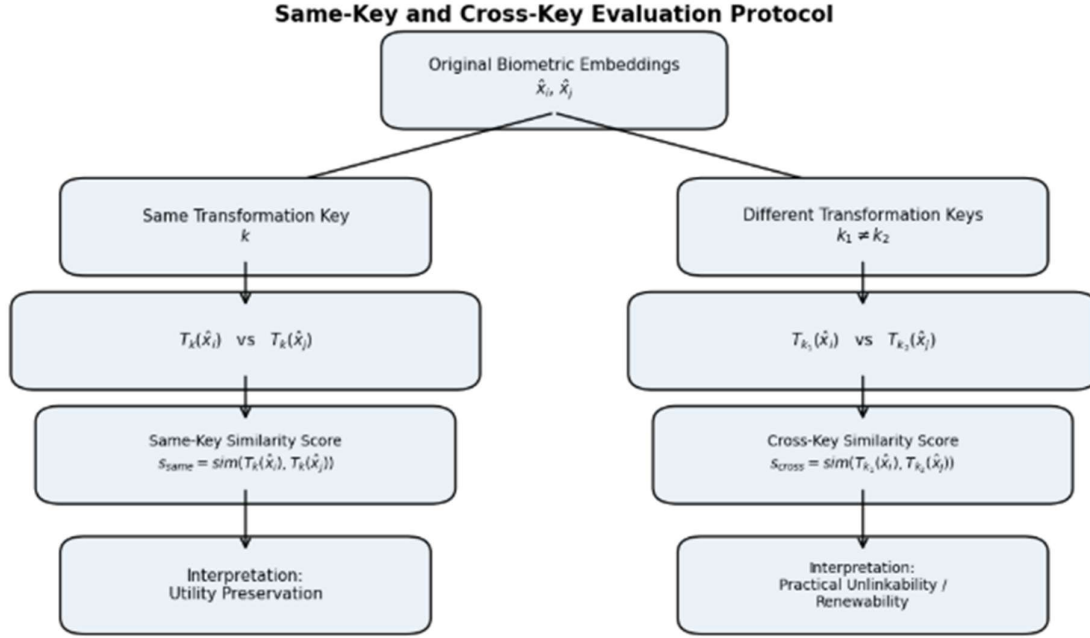


Fig. 4.2. Same-key and cross-key evaluation protocol for analysing protected-template utility and practical unlinkability.

This distinction is central to the methodology. Same-key performance alone is not sufficient because a protected template may remain linkable across different keys. Cross-key unlinkability alone is also not sufficient if same-key performance becomes too weak for practical verification. Therefore, both evaluations are required to interpret the security–utility balance of the protected system.

Cross-key evaluation is interpreted as empirical evidence of practical unlinkability under the tested protocol. It is not claimed as a formal proof of irreversibility or unconditional unlinkability.

4.9 Protected Score-Level Fusion

After protected unimodal scores are obtained, score-level fusion is applied to evaluate whether multimodal fusion can recover the utility loss introduced by cancellable transformation. Fusion

is especially important in protected biometric systems because template protection may affect each modality differently.

The protected score of modality r is denoted by S_r . For a bimodal system, the fused protected score is computed as:

$$S_f = w_1 S_1 + w_2 S_2$$

For a trimodal system, the fused protected score is computed as:

$$S_f = w_1 S_1 + w_2 S_2 + w_3 S_3$$

where w_1 , w_2 , and w_3 are fusion weights selected using the validation set.

The following protected fusion systems are evaluated:

Table 4.2 Protected Fusion Systems Evaluated

Fusion System	Modalities Used
Face + Fingerprint	Face and fingerprint protected scores
Face + Iris	Face and iris protected scores
Fingerprint + Iris	Fingerprint and iris protected scores
Face + Fingerprint + Iris	Face, fingerprint, and iris protected scores

The analytical role of fusion in this work is important. Fusion is not only used to improve accuracy. It is used to test whether the protected system can recover utility after cancellable transformation. Since different modalities lose discriminative information differently, their errors may not fully overlap. Therefore, fusion can combine complementary protected evidence and reduce dependence on a single modality [7]–[10].

The reason fusion can recover protected performance is that cancellable transformation affects modalities differently. Iris may retain stronger discriminative structure, face may provide moderate complementary evidence, and fingerprint may contribute auxiliary information even when its unimodal protected performance is weaker. Since the modality errors are not fully identical, score-level fusion can reduce the effect of degradation in any single protected matcher.

4.10 CKKS Encrypted Matching on Protected Templates

CKKS encrypted matching is applied after cancellable transformation to evaluate computation-level privacy in the protected domain. This stage examines whether similarity computation on protected templates can be carried out in encrypted form while preserving protected-domain verification behaviour.

For a protected template $T_k(\hat{x})$, CKKS encryption produces an encrypted representation:

$$Enc(T_k(\hat{x}))$$

The encrypted similarity computation follows the same inner-product structure as protected plain matching:

$$Enc(s) = Enc(T_k(\hat{x}_i)) \cdot T_k(\hat{x}_j)$$

where one protected template is encrypted and the comparison template is used in plaintext form for encrypted inner-product computation. The decrypted score corresponds to the protected-domain similarity score.

After decryption, the resulting score is compared with the protected-domain plain score. If the scores and verification metrics remain equivalent, then CKKS encrypted matching preserves protected-domain matching behaviour.

CKKS is suitable for this task because it supports approximate arithmetic over real-valued vectors [14]. TenSEAL provides practical tools for implementing CKKS encrypted tensor operations [15].

The CKKS configuration used in the protected matching stage is consistent with the baseline setting. This stage is important because it separates recognition utility from computation privacy. If protected CKKS matching gives the same verification behaviour as protected plain matching, then the main cost of CKKS is computational overhead rather than recognition degradation.

4.11 Formal Security–Utility Evaluation Strategy

The proposed framework is evaluated using a formal security–utility strategy rather than an accuracy-only strategy. This is necessary because privacy-preserving biometric systems must be judged by both recognition performance and protection behaviour.

The evaluation considers the following dimensions:

1. **Protected utility:** measured using same-key EER, ROC-AUC, accuracy, and TAR@FAR.
2. **Practical unlinkability:** analysed using cross-key score distributions and cross-key verification behaviour.
3. **Fusion recovery:** measured by comparing protected unimodal performance with protected bimodal and trimodal fusion.
4. **Encrypted matching preservation:** measured by comparing protected plain results with protected CKKS results.
5. **Runtime overhead:** measured using encryption time, matching time, and per-pair latency.

The security–utility trade-off is interpreted as follows. A strong cancellable transformation may improve renewability and reduce linkability, but it may also distort identity information and reduce same-key performance. A weak transformation may preserve utility but may provide weaker unlinkability. Therefore, the best configuration is not necessarily the one with the lowest EER alone. It is the configuration that provides acceptable same-key performance, weak cross-key matchability, strong fusion recovery, and feasible runtime.

This evaluation strategy directly supports the system-level contribution of the work. It treats secure biometric verification as a balance between template protection, verification utility, unlinkability, encrypted computation, and computational cost.

4.12 Research Questions and Evaluation Evidence

To make the methodology analytically clear, Table 4.4 maps each research question to the evaluation evidence used in this project.

Table 4.3 Research Questions and Evaluation Evidence

Research Question	Evaluation Evidence
Does cancellable protection preserve utility?	Same-key EER, ROC-AUC, accuracy, TAR@FAR
Does key renewal reduce linkability?	Cross-key score distributions and cross-key verification behaviour
Does fusion recover protected performance?	Protected unimodal vs. protected bimodal/trimodal EER and ROC-AUC
Does CKKS change recognition behaviour?	Protected plain vs. protected CKKS comparison
What is the computational cost?	Encryption time, matching time, per-pair latency
What is the best operating balance?	Security–utility and latency-over-security analysis

This mapping ensures that each methodological component is connected to a specific evaluation purpose. It also makes the contribution clearer by showing that the framework is not only an implementation pipeline, but a structured research evaluation.

4.13 Summary

This chapter presented the methodology of Major Project-II. The proposed framework extends the Major Project-I baseline by adding cancellable template transformation, binary and non-binary protected templates, projection dimension analysis, same-key and cross-key verification, protected score-level fusion, and CKKS encrypted matching over protected templates.

The methodology is designed as a layered privacy-preserving framework. Cancellable transformation provides template-level renewability, same-key evaluation measures retained utility, cross-key evaluation provides empirical evidence of practical unlinkability, protected fusion evaluates utility recovery, and CKKS encrypted matching provides computation-level privacy.

The chapter also formally defined the threat model and security assumptions. The system is evaluated under an honest-but-curious server setting, with attention to template compromise, cross-key linkage, protected computation, and runtime overhead. The evaluation strategy focuses on the complete security–utility balance rather than recognition accuracy alone.

The next chapter presents the experimental setup and evaluation protocol used to implement this methodology and compare plain, protected, fused, and encrypted verification systems.

CHAPTER 5

Experimental Setup And Evaluation Protocol

5.1 Introduction

This chapter presents the experimental setup and evaluation protocol used to validate the proposed privacy-preserving multimodal biometric verification framework. The purpose of the experimental design is not only to report recognition performance but also to analyse the complete security–utility behaviour of the system.

The experiments are designed to answer five main questions. First, whether cancellable transformation preserves sufficient verification utility under same-key matching. Second, whether cross-key matching provides empirical evidence of practical unlinkability and renewability. Third, whether protected multimodal fusion can recover performance loss introduced by template protection. Fourth, whether CKKS encrypted matching preserves protected-domain verification behaviour. Fifth, what computational overhead is introduced by encrypted matching and protected fusion.

To ensure fair comparison, the same subject-level split, balanced verification-pair protocol, embedding pipeline, and evaluation metrics are used across plain, protected, fused, and CKKS encrypted settings. This controlled protocol is important because the proposed contribution is system-level. Therefore, differences in performance must reflect the effect of template protection, fusion, encryption, and runtime overhead rather than differences in data split or pair generation.

5.2 Dataset and Split Protocol

The experiments use a virtually subject-aligned multimodal dataset constructed from three public biometric sources: VGGFace2 for face, SOCOFing for fingerprint, and CASIA-Iris-Thousand for iris [26]–[28]. A virtual subject-alignment strategy is used because a large public

dataset containing naturally aligned face, fingerprint, and iris samples from the same individuals was not available for the required experimental setting.

The dataset contains 600 virtual subjects. Each subject has 10 samples per modality. Since three modalities are used, each subject contains 30 biometric samples. Therefore, the complete dataset contains 18,000 biometric samples.

$$600 \times 10 \times 3 = 18,000$$

A subject-level split is used to avoid identity leakage between training, validation, and testing. This means that subjects used for training are not repeated in validation or testing. The split is shown in Table 5.1.

Table 5.1 Experimental Data Protocol

Component	Description
Modalities	Face, fingerprint, iris
Source datasets	VGGFace2, SOCOFing, CASIA-Iris-Thousand
Number of virtual subjects	600
Samples per modality per subject	10
Total biometric samples	18,000
Training subjects	480
Validation subjects	60
Testing subjects	60

The validation set is used for score normalization and fusion-weight selection. The test set is used only for final performance reporting. This separation prevents the fusion weights or thresholds from being optimized directly on the final test data.

5.3 Verification Pair Generation

The verification task is formulated as a pairwise matching problem. Each comparison pair is labelled as either genuine or impostor. A genuine pair contains two samples belonging to the same subject, while an impostor pair contains samples belonging to different subjects.

For both validation and testing, balanced verification-pair sets are generated. Each split contains 2,700 genuine pairs and 2,700 impostor pairs, giving a total of 5,400 pairs per modality.

The same balanced pair protocol is used across all modalities and all evaluation settings. This is important for system-level comparison because it ensures that unimodal, multimodal, protected, and encrypted systems are evaluated under the same matching conditions.

The balanced pair protocol also supports clearer interpretation of EER, accuracy, ROC-AUC, and TAR@FAR. Since the number of genuine and impostor comparisons is equal, the results are not biased by a large imbalance between positive and negative pairs.

5.4 Embedding Preparation

Deep embeddings are extracted separately for face, fingerprint, and iris. Face embeddings are obtained using an InceptionResnetV1 model pretrained on VGGFace2. Fingerprint embeddings are extracted using a ResNet-based fingerprint model. Iris embeddings are extracted using an EfficientNet-B3-based model.

Each modality produces a 512-dimensional embedding representation. These embeddings are normalized before matching and before cancellable transformation. Normalization provides stable vector magnitudes and supports cosine similarity or normalized dot-product matching.

The same embeddings are used across plain, protected, fused, and CKKS encrypted experiments. This ensures that performance differences are caused by the evaluation configuration rather than by different feature representations.

5.5 Experimental Configurations

The experimental design compares multiple verification settings to evaluate the contribution of each privacy-preserving layer. The configurations include plain-domain matching, baseline CKKS matching, cancellable protected matching, protected score-level fusion, and protected CKKS matching.

Table 5.2 Experimental Configurations

Configuration	Description	Purpose
Plain unimodal matching	Direct matching of normalized embeddings	Establish unprotected baseline utility

Plain score-level fusion	Fusion of plain modality scores	Measure multimodal performance before protection
CKKS baseline matching	Encrypted matching on baseline embeddings	Evaluate computation-level privacy
Protected unimodal matching	Matching after cancellable transformation	Measure protected-template utility
Same-key evaluation	Protected templates compared using same key	Analyse retained utility
Cross-key evaluation	Protected templates compared using different keys	Analyse practical unlinkability
Protected fusion	Fusion of protected modality scores	Measure utility recovery after protection
Protected CKKS matching	CKKS matching over protected templates	Evaluate encrypted protected-domain matching
Runtime analysis	Encryption and matching time measurement	Analyse computational overhead

This configuration structure is important because it separates different effects. Plain matching shows the original recognition capability. Protected matching shows the effect of cancellable transformation. Cross-key matching evaluates unlinkability. Protected fusion measures recovery after protection. CKKS matching evaluates whether encrypted computation changes matching behaviour. Runtime analysis measures practical cost.

5.6 Cancellable Protection Settings

Cancellable biometric protection is implemented using key-dependent random projection. The original normalized embedding is transformed into a protected representation using a transformation key. Different keys generate different protected templates from the same original embedding, supporting renewability and diversity [11]–[13].

Both binary and non-binary protected templates are evaluated. The non-binary protected template retains continuous projected values, while the binary protected template applies sign-based binarization. These two variants are included because they represent different security–utility behaviours. Non-binary templates may preserve more matching information, while binary templates provide a more compact representation but may lose discriminative detail.

The following projection dimensions are evaluated:

$$m \in \{64, 128, 256, 512, 1024\}$$

Table 5.3 Cancellable Protection Settings

Parameter	Values / Description
Transformation method	Key-dependent random projection
Original embedding dimension	512
Protected dimensions	64, 128, 256, 512, 1024
Template variants	Binary and non-binary
Evaluation keys	Same-key and cross-key settings
Purpose	Utility, renewability, unlinkability, security-utility trade-off

These settings allow the experiment to analyse whether stronger compression or binarization improves protection at the cost of verification utility. The evaluation is performed separately for face, fingerprint, and iris to identify modality-dependent behaviour.

5.7 Same-Key and Cross-Key Evaluation Protocol

Same-key and cross-key evaluation are used to formally analyse cancellable template behaviour.

In the same-key setting, both templates in a pair are generated using the same transformation key. Same-key verification measures protected-template utility. A useful protected system should preserve high similarity for genuine pairs and low similarity for impostor pairs.

In the cross-key setting, templates are generated using different transformation keys. Cross-key verification evaluates whether renewed templates become difficult to link. If cross-key genuine and impostor score distributions overlap strongly, this provides empirical evidence of practical unlinkability [5], [6].

Table 5.4 Same-Key and Cross-Key Evaluation Protocol

Evaluation Setting	Key Condition	Main Purpose	Expected Behaviour
Same-key	Same key used for both templates	Utility preservation	Genuine and impostor scores remain separable
Cross-key	Different keys used for the two templates	Practical unlinkability and renewability	Genuine and impostor scores become highly overlapping

This evaluation is important because same-key accuracy alone is not sufficient for a secure cancellable biometric system. A system may verify users correctly but still allow templates to

be linked across keys. Therefore, same-key and cross-key results are interpreted together as part of the security–utility analysis.

Cross-key results are treated as empirical evidence of practical unlinkability under the tested protocol. They are not claimed as a formal mathematical proof of irreversibility or unconditional unlinkability.

5.8 Protected Fusion and Weight Selection

Protected score-level fusion is evaluated to determine whether multimodal fusion can recover the performance loss introduced by cancellable transformation. This is a central part of the proposed system-level contribution because fusion is evaluated not only for accuracy improvement but also as a utility-recovery mechanism after template protection.

Before fusion, protected modality scores are normalized to make them comparable. Fusion weights are selected using the validation set and then applied to the test set. This prevents the final test performance from being directly optimized.

For a bimodal protected fusion system, the fused score is computed as:

$$S_f = w_1 S_1 + w_2 S_2$$

For the trimodal system, the fused score is computed as:

$$S_f = w_1 S_1 + w_2 S_2 + w_3 S_3$$

where S_1 , S_2 , and S_3 represent protected modality scores, and w_1 , w_2 , and w_3 represent validation-selected fusion weights.

Fusion recovery is expected because different modalities are affected differently by cancellable transformation. If their errors are not fully identical, fused scores can reduce the impact of a weak protected modality and recover part of the utility lost after protection [7]–[10].

5.9 CKKS Encrypted Matching Settings

CKKS encrypted matching is evaluated to determine whether protected-domain similarity computation can be performed in encrypted form without changing verification behaviour. CKKS is selected because it supports approximate arithmetic over real-valued vectors [14]. TenSEAL is used for practical implementation of encrypted tensor operations [15]. The CKKS parameters used in the experiments are shown in Table 5.8.

Table 5.5 CKKS Parameter Settings

Parameter	Value
Scheme	CKKS
poly_modulus_degree	8192
coeff_mod_bit_sizes	[60, 40, 40, 60]
global_scale	2^{40}
Operation	Encrypted inner product / similarity computation
Evaluation setting	Protected CKKS matching

In this evaluation, one protected template is encrypted and the comparison template is used in plaintext form for encrypted inner-product computation. The decrypted score is compared with the protected plain-domain score. If the metrics remain unchanged or nearly unchanged, CKKS is interpreted as preserving protected-domain verification behaviour.

The purpose of this experiment is to separate recognition loss from computational overhead. If CKKS produces the same verification behaviour as protected plain matching, then the main effect of encryption is runtime cost rather than accuracy degradation.

5.10 Evaluation Metrics

The proposed framework is evaluated using recognition metrics, security-oriented metrics, fusion-recovery metrics, and runtime metrics. This is necessary because privacy-preserving biometric systems cannot be evaluated by accuracy alone.

The primary biometric metrics used are Equal Error Rate, ROC-AUC, accuracy, and True Acceptance Rate at fixed False Acceptance Rate.

Equal Error Rate (EER): EER is the point at which the false acceptance rate and false rejection rate are equal. A lower EER indicates better verification performance.

$$FAR = FRR$$

ROC-AUC: ROC-AUC measures the separability between genuine and impostor score distributions. A higher ROC-AUC indicates stronger discriminative performance.

Accuracy: Accuracy measures the proportion of correctly classified genuine and impostor pairs.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

TAR@FAR: TAR@FAR measures the true acceptance rate at a fixed false acceptance rate. This metric is important because biometric systems are often evaluated under low-FAR operating conditions.

$$TAR = 1 - FRR$$

Recovery Gain: Recovery gain measures how much performance is restored by protected multimodal fusion compared with the best protected unimodal system.

$$Absolute\ Gain = EER_{best\ protected\ unimodal} - EER_{protected\ fusion}$$

$$Relative\ Gain(\%) = \frac{EER_{best\ protected\ unimodal} - EER_{protected\ fusion}}{EER_{best\ protected\ unimodal}} \times 100$$

Runtime Metrics: Runtime is evaluated using encryption time, matching time, and per-pair latency. This helps analyse the practical cost of CKKS encrypted matching.

Table 5.6 Evaluation Metrics

Metric	Purpose
EER	Measures verification error at balanced FAR and FRR
ROC-AUC	Measures genuine/impostor score separability
Accuracy	Measures overall classification correctness
TAR@FAR	Measures performance at low false-acceptance rates
Cross-key score behaviour	Evaluates practical unlinkability
Recovery gain	Measures fusion-based utility recovery
Encryption time	Measures CKKS encryption overhead
Matching time	Measures protected and encrypted matching cost
Per-pair latency	Measures practical matching efficiency

These metrics collectively support the system-level security–utility interpretation of the proposed framework.

5.11 Recovery Gain and Runtime Evaluation

Recovery gain is used to quantify the role of protected fusion after cancellable transformation. Since template protection may reduce unimodal performance, recovery gain measures whether bimodal or trimodal fusion restores verification utility.

The recovery gain analysis compares the best protected unimodal system with each protected fusion system. If the fused system achieves lower EER than the best protected unimodal system, the reduction is interpreted as fusion-based utility recovery. This supports the claim that fusion is not only an accuracy-enhancement step but also a recovery mechanism after protection.

Runtime evaluation is performed to measure the computational cost of privacy-preserving matching. The main runtime components are encryption time, encrypted matching time, and per-pair latency. This analysis is important because a system with strong security and high accuracy may still be impractical if its computational overhead is too high.

The runtime analysis also supports the latency-over-security discussion. Plain protected matching provides faster computation, while protected CKKS matching provides stronger computation-level privacy at higher cost. The trimodal protected CKKS setting gives the strongest layered protection but may introduce additional runtime overhead.

5.12 Reporting Strategy

The experimental results are reported in a structured manner to support clear interpretation. First, plain and CKKS baseline results are reported to establish reference performance. Second, protected unimodal results are reported to show the effect of cancellable transformation on each modality. Third, same-key and cross-key score behaviour is analysed to interpret utility and practical unlinkability. Fourth, protected fusion results are reported to evaluate recovery after

template protection. Fifth, protected CKKS results are compared with protected plain results to analyse encrypted matching preservation. Finally, runtime and latency-over-security results are reported to discuss practical deployment trade-offs.

This reporting strategy is aligned with the system-level contribution of the project. It avoids treating accuracy as the only outcome and instead evaluates recognition performance, template renewability, unlinkability, protected fusion recovery, encrypted computation, and computational cost together.

5.13 Summary

This chapter presented the experimental setup and evaluation protocol used for the proposed privacy-preserving multimodal biometric verification framework. The experiments use a virtually subject-aligned face–fingerprint–iris dataset with 600 subjects, 10 samples per modality, and 18,000 total biometric samples. A subject-level split and balanced verification-pair protocol are used to ensure fair evaluation.

The chapter described embedding preparation, experimental configurations, cancellable protection settings, same-key and cross-key evaluation, protected fusion and weight selection, CKKS encrypted matching, evaluation metrics, recovery gain, runtime evaluation, and reporting strategy.

The experimental design supports the main research contribution of the work. It evaluates the system not only through recognition accuracy but also through protected-template utility, practical unlinkability, fusion-based recovery, encrypted matching preservation, and computational overhead. The next chapter presents the results and analysis obtained using this evaluation protocol.

CHAPTER 6

Result And Analysis

6.1 Introduction

This chapter presents the results and analysis of the proposed privacy-preserving multimodal biometric verification framework. The evaluation is not limited to reporting recognition accuracy. Instead, the results are interpreted from a system-level security–utility perspective, where verification performance, template renewability, cross-key unlinkability, protected fusion recovery, encrypted computation, and runtime overhead are analysed together.

The results are discussed in six stages. First, the plain-domain and CKKS unimodal baseline results are presented to establish the reference behaviour before cancellable template protection. Second, protected unimodal verification results are analysed to study the effect of cancellable transformation on face, fingerprint, and iris modalities. Third, same-key and cross-key score behaviours are interpreted to evaluate protected-template utility and practical unlinkability. Fourth, protected multimodal fusion results are analysed to determine whether fusion can recover utility loss introduced by template protection. Fifth, protected CKKS matching results are compared with protected plain matching to examine whether encrypted computation preserves protected-domain verification behaviour. Finally, recovery gain, latency, and security–utility trade-off are discussed.

The purpose of this chapter is not only to identify the best-performing configuration, but also to interpret the role of each privacy-preserving layer. Plain and CKKS baseline results establish the reference behaviour. Protected unimodal results show the effect of cancellable transformation. Same-key and cross-key results separate utility preservation from practical unlinkability. Protected fusion results test whether complementary modalities can recover utility after transformation. Protected CKKS results determine whether encrypted computation

changes the protected-domain behaviour. Therefore, the analysis is structured to establish a system-level insight rather than only report performance numbers.

A privacy-preserving biometric system cannot be evaluated only by EER or accuracy. It must also be analysed in terms of template renewability, unlinkability, fusion-based recovery, encrypted computation, and computational cost [3]–[6], [11]–[15].

6.2 Baseline Plain and CKKS Unimodal Verification Results

The first experiment evaluates the plain-domain and CKKS encrypted-domain performance of face, fingerprint, and iris verification. These results provide the baseline reference before cancellable template transformation is applied.

Table 6.1 Unimodal Plain and CKKS Verification Results

Modality	Plain EER	Plain ROC-AUC	CKKS EER	CKKS ROC-AUC	Interpretation
Face	5.31%	0.9865	5.31%	0.9865	Strong baseline performance
Fingerprint	16.30%	0.9119	16.30%	0.9119	More challenging modality
Iris	1.80%	0.9983	1.80%	0.9983	Strongest unimodal performance

The baseline results show that iris provides the strongest unimodal verification performance, followed by face, while fingerprint shows comparatively weaker performance. Iris achieves the lowest EER and the highest ROC-AUC, which indicates strong separability between genuine and impostor scores. Face also provides strong baseline performance, whereas fingerprint is more challenging because fingerprint matching may be affected by intra-class variation, image quality, partial impressions, and local ridge-pattern variations.

The CKKS encrypted results preserve the plain-domain verification behaviour. The EER and ROC-AUC values remain unchanged in the reported setting. This indicates that CKKS encrypted matching does not materially affect recognition performance when the encrypted computation follows the same inner-product structure as the plain-domain similarity computation [14], [15].

The significance of this result is that CKKS can provide computation-level privacy without introducing recognition degradation in the tested setting. However, this does not mean that CKKS alone provides complete biometric template protection. CKKS protects the computation stage, but it does not directly provide template revocability, renewability, or cross-key unlinkability. Therefore, the baseline results support the need for the cancellable template

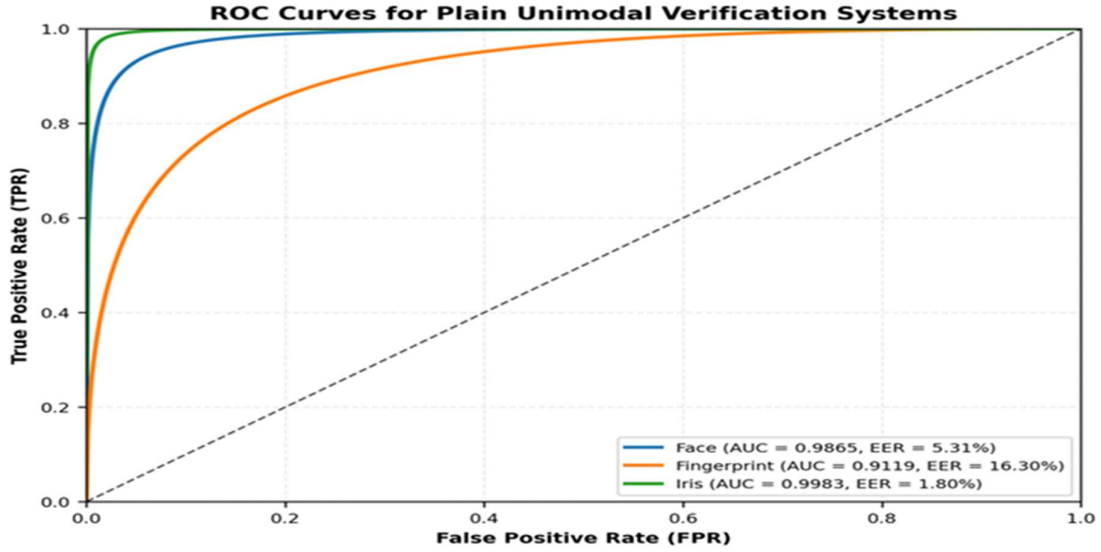


Fig. 6.1. ROC curves for plain-domain unimodal verification using face, fingerprint, and iris modalities.

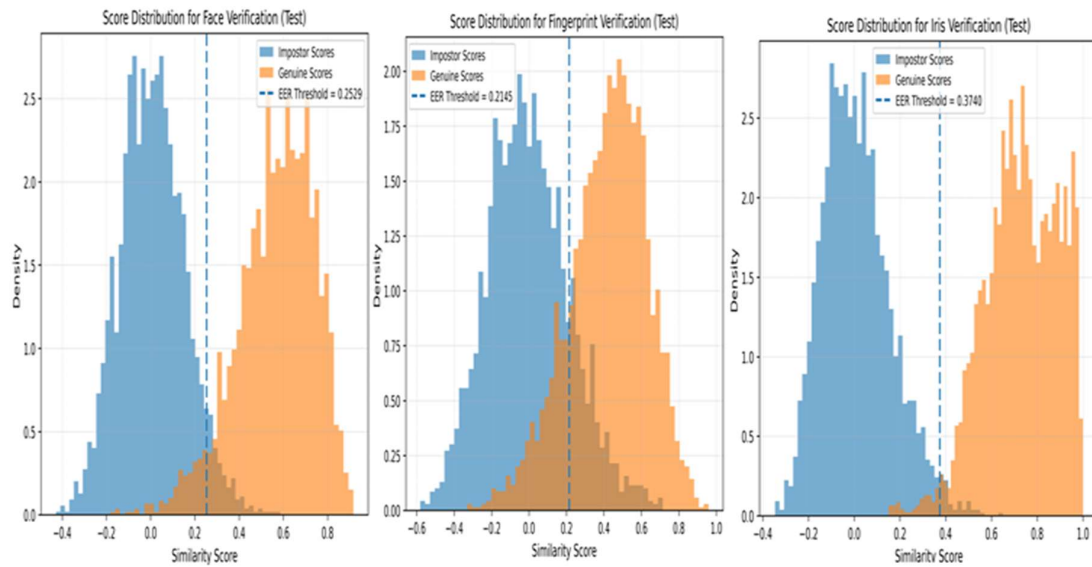


Fig. 6.2. Genuine and impostor score distributions for unimodal baseline systems.

6.3 Protected Unimodal Verification Results

The next experiment evaluates the effect of cancellable template transformation on each unimodal biometric system. The objective is to measure whether protected templates retain sufficient verification utility under same-key matching.

Table 6.2 Best Protected Unimodal Configurations

Modality	Best Protected EER	ROC-AUC	Interpretation
Face	5.74%	0.9849	Moderate degradation after protection
Fingerprint	18.43%	0.8950	Most sensitive to transformation
Iris	2.04%	0.9979	Strongest protected unimodal system

The protected unimodal results show that cancellable transformation affects the three modalities differently. Iris retains the strongest protected-domain performance, with only a small degradation compared with the plain-domain baseline. Face also remains effective, although its EER increases moderately after protection. Fingerprint shows the largest degradation, indicating that its embedding representation is more sensitive to projection and binarization effects.

This modality-dependent behaviour is important for the research contribution of the work. It shows that protected-template configuration should not be selected using a single fixed assumption for all modalities. Instead, each modality must be evaluated separately because cancellable transformation may preserve or distort discriminative information differently.

The result also confirms the security–utility trade-off in cancellable biometrics. The transformation provides revocability and key-dependent protection, but it may alter the embedding geometry that supports matching [11]–[13]. Therefore, protected-template performance must be interpreted together with cross-key unlinkability and fusion recovery rather than only through unimodal EER.

6.4 Same-Key Protected Score Distribution Analysis

Same-key evaluation measures whether protected templates generated using the same transformation key retain enough discriminative information for verification. In this setting, genuine pairs should produce higher similarity scores than impostor pairs. A clear separation between genuine and impostor score distributions indicates good protected-template utility.

The same-key results show that protected templates remain usable for verification, especially for iris and face. Iris maintains strong score separation, which explains its low protected EER. Face shows moderate overlap between genuine and impostor distributions but still retains useful verification performance. Fingerprint shows greater overlap, which explains its weaker protected-domain performance.

This result shows that cancellable transformation does not affect all modalities uniformly. A useful protected biometric framework must therefore consider modality-specific behaviour. Iris can serve as a strong protected modality, face provides additional reliable evidence, and fingerprint may still contribute complementary information in fusion even if its unimodal protected performance is weaker.

Same-key evaluation should therefore be interpreted as a measure of protected-template utility. It answers the question: after applying cancellable transformation, does the protected representation still preserve enough identity-discriminative information for verification? The results indicate that the answer is yes, but the degree of utility preservation depends on the modality.

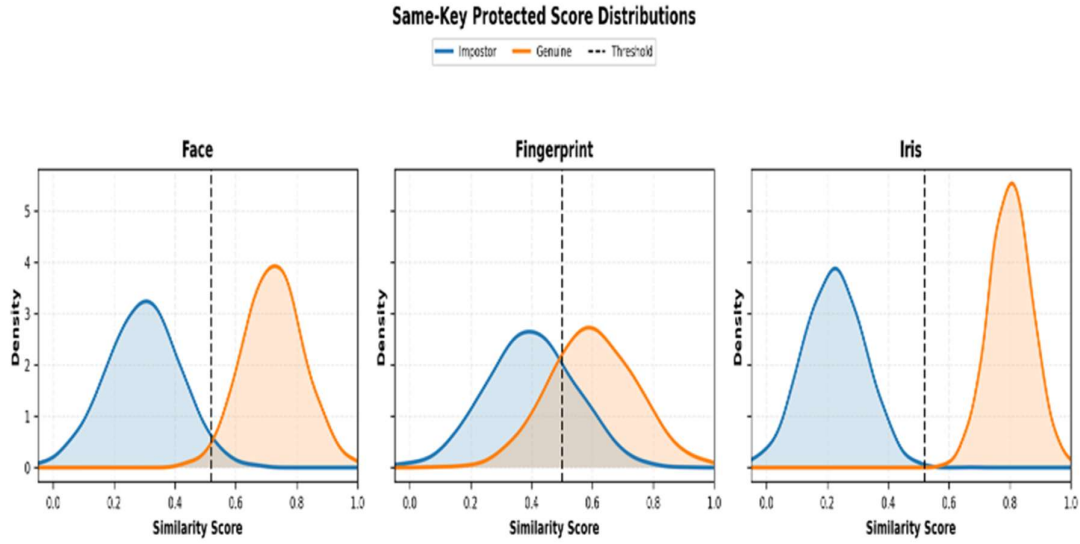


Fig. 6.3. Same-key protected score distributions for the best protected unimodal configurations.

6.5 Cross-Key Protected Score Distribution Analysis

Cross-key evaluation measures whether protected templates generated using different transformation keys become difficult to link. In this setting, even genuine samples from the same subject should not remain strongly matchable if the transformation is key-dependent and renewable.

The cross-key score distributions show substantially higher overlap between genuine and impostor scores compared with same-key evaluation. This behaviour supports practical unlinkability because templates generated under different keys do not preserve the same matching relationship as same-key templates. In other words, a renewed template generated using a different key becomes difficult to associate with the earlier protected template.

This result should be interpreted carefully. Cross-key behaviour provides empirical evidence of practical unlinkability under the tested protocol, but it is not a formal mathematical proof of irreversibility or unconditional unlinkability [5], [6]. The value of the cross-key experiment is

that it tests a practical attack scenario: whether protected templates generated from the same biometric source but under different keys remain linkable.

The combination of same-key and cross-key results is central to the security–utility interpretation. Same-key evaluation shows retained utility, while cross-key evaluation shows reduced linkability after renewal. A strong cancellable biometric system should show both acceptable same-key verification and weak cross-key matchability. Therefore, these two evaluations should not be treated separately; they jointly define the practical usefulness of the protected template.

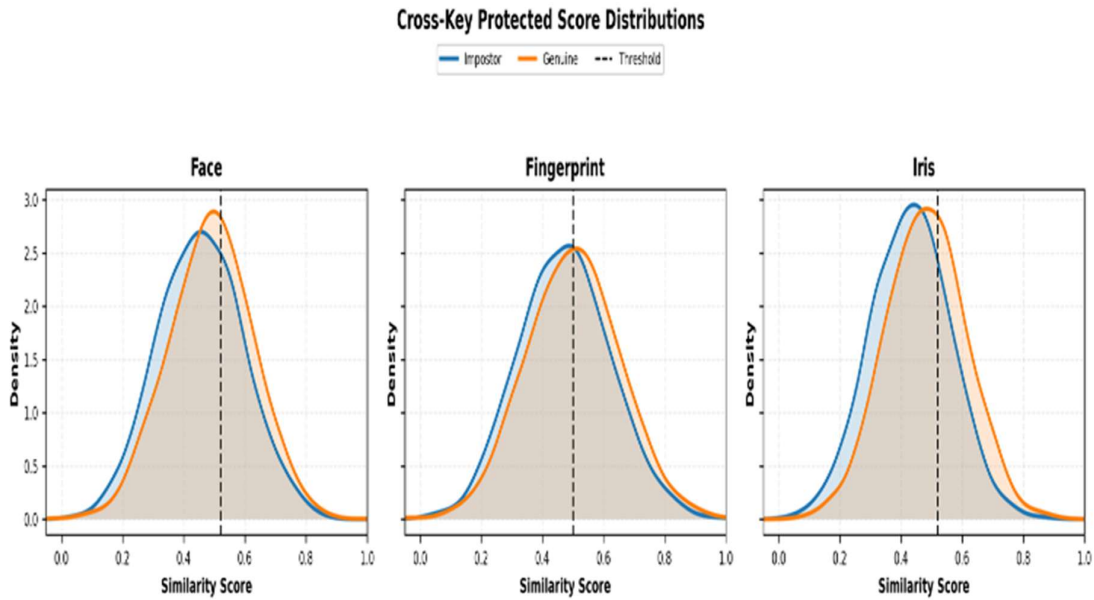


Fig. 6.4. Cross-key protected score distributions for analysing practical unlinkability and renewability.

6.6 Protected Multimodal Fusion Results

Protected multimodal fusion is evaluated to determine whether score-level fusion can recover the verification loss introduced by cancellable template transformation. The purpose of fusion in this stage is not only to improve accuracy but also to test whether complementary biometric evidence remains useful after template protection.

Table 6.3 Fusion Performance under Plain, Protected, and Protected CKKS Settings

Fusion System	Plain EER	Protected EER	Protected + CKKS EER	Protected + CKKS ROC-AUC
Face + Fingerprint	3.52%	3.48%	3.48%	0.9935
Face + Iris	0.33%	0.41%	0.41%	0.9999
Fingerprint + Iris	1.15%	1.41%	1.41%	0.9991
Face + Fingerprint + Iris	0.33%	0.31%	0.31%	0.9999

The protected fusion results show that multimodal fusion significantly improves performance compared with protected unimodal systems. The protected trimodal system achieves the strongest performance, with 0.31% EER and 0.9999 ROC-AUC. This is substantially better than the best protected unimodal result.

The improvement occurs because cancellable transformation affects each modality differently. Iris retains strong discriminative structure, face provides additional identity evidence, and fingerprint contributes auxiliary information even when its unimodal performance is weaker. The errors produced by the protected unimodal systems are therefore not fully identical. Score-level fusion combines partially complementary protected evidence and reduces dependence on any single protected matcher [7]–[10].

This establishes a central research insight of the work: protected multimodal fusion acts as a utility-recovery layer after cancellable transformation. The improvement is not only because more scores are added, but because the protected modalities preserve different portions of identity-discriminative information. Since the modality-specific errors are partially non-overlapping, score-level fusion reduces the effect of degradation in any single modality. Thus, fusion becomes a necessary component of the protected system rather than an optional accuracy-enhancement step.

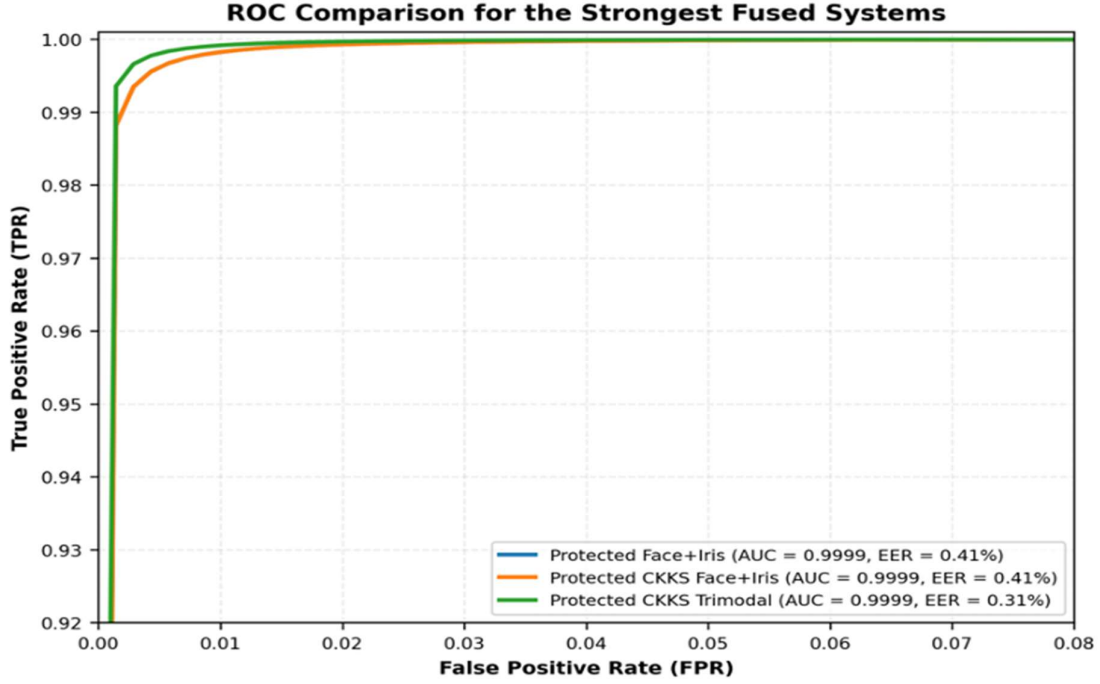


Fig. 6.5. ROC comparison of protected multimodal fusion systems.

6.7 Protected CKKS Matching Results

Protected CKKS matching evaluates whether encrypted-domain similarity computation preserves protected-domain verification behaviour. In this experiment, CKKS encrypted matching is applied after cancellable transformation.

The results show that protected CKKS matching produces the same verification behaviour as protected plain matching for the evaluated systems. The EER and ROC-AUC values remain unchanged in the reported settings. This indicates that CKKS encryption does not introduce additional recognition loss when the encrypted computation follows the same protected similarity structure [14], [15].

This result has an important interpretation. The recognition degradation observed in protected systems is caused mainly by cancellable transformation, not by CKKS encrypted computation. CKKS adds computation-level privacy, but its main cost is runtime overhead rather than accuracy degradation.

This result also separates the roles of the two privacy mechanisms. Cancellable transformation changes the template representation and therefore affects recognition utility, renewability, and unlinkability. CKKS, in contrast, preserves the protected similarity computation and mainly affects runtime. Therefore, the layered framework should be interpreted as template-level protection followed by computation-level privacy, rather than as two interchangeable security mechanisms.

6.8 Recovery Gain after Template Protection

Recovery gain is used to measure how much performance is restored by multimodal fusion after cancellable template transformation. Since template protection may distort the original embedding space, protected unimodal systems may show higher EER than their plain-domain counterparts. Recovery gain therefore provides a system-level measure of whether fusion can compensate for this degradation.

Table 6.4 Recovery Gain after Template Protection

System	EER	Absolute Gain over Best Protected Unimodal	Relative Gain
Best Protected Unimodal: Iris	2.04%	0.00%	0.00%
Protected Face + Fingerprint	3.48%	-1.44%	Negative gain
Protected Face + Iris	0.41%	1.63%	79.90%
Protected Fingerprint + Iris	1.41%	0.63%	30.88%
Protected Face + Fingerprint + Iris	0.31%	1.73%	84.80%

The protected trimodal system provides the strongest recovery gain. Compared with the best protected unimodal system, it reduces EER from 2.04% to 0.31%. This corresponds to an absolute gain of 1.73% and a relative recovery gain of approximately 84.80%.

This gain is not only a numerical improvement. It shows that different protected modalities retain complementary identity information after cancellable transformation. Iris contributes the strongest protected signal, face adds additional discriminative evidence, and fingerprint

provides auxiliary information. When these protected scores are normalized and fused, the final system can recover a significant portion of the utility loss introduced by template protection.

The negative gain in the Face + Fingerprint system shows that fusion is not automatically beneficial in every combination. Fusion becomes effective when the modalities provide sufficiently strong and complementary evidence. Since iris remains highly discriminative after protection, fusion systems involving iris provide stronger recovery. This reinforces the importance of modality-aware interpretation rather than assuming that all multimodal combinations will behave equally.

This supports one of the central conclusions of the work: multimodal fusion is an important design component for privacy-preserving biometric systems because it can restore verification utility while preserving the benefits of template protection.

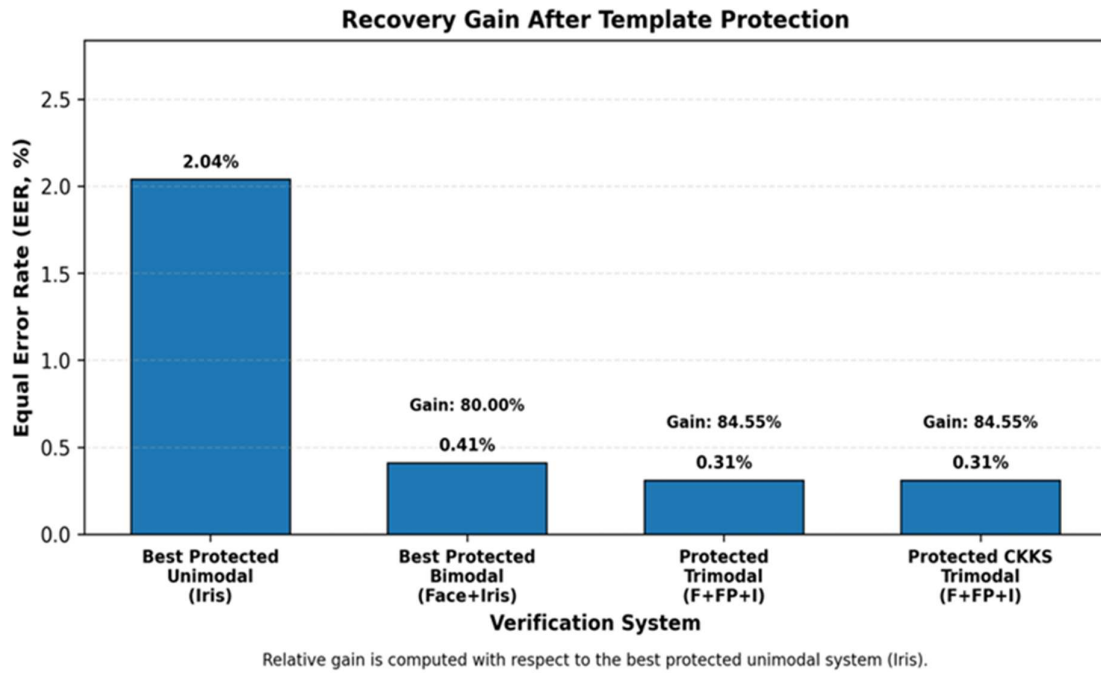


Fig. 6.6. Recovery gain after template protection for protected unimodal and fusion systems.

6.9 Latency-over-Security Analysis

Runtime analysis is performed to evaluate the computational cost of the proposed privacy-preserving framework. This is important because a biometric system must be both secure and practically usable.

Table 6.5 Latency-over-Security Operating Modes

Operating Mode	Protection Level	Matching Setting	Expected Advantage	Expected Cost
Fast Protected Mode	Cancellable template only	Protected plain matching	Faster matching and template renewability	No encrypted computation
Balanced Protected CKKS Mode	Cancellable + CKKS	Bimodal protected CKKS	Stronger privacy with moderate cost	Higher runtime than plain protected matching
High-Security Trimodal CKKS Mode	Cancellable + CKKS + trimodal fusion	Trimodal protected CKKS	Strongest layered protection and best utility	Highest computational overhead

The latency-over-security analysis shows that different deployment settings may require different operating modes. If speed is the main requirement, protected plain matching provides template renewability with lower computational cost. If computation-level privacy is required, protected CKKS matching provides stronger privacy but increases runtime. If the strongest utility and layered protection are required, trimodal protected CKKS matching provides the best security–utility behaviour but with higher computational overhead.

This analysis reinforces the system-level nature of the work. The best configuration is not necessarily the fastest or the most accurate alone. It depends on the required balance between utility, unlinkability, encrypted computation, and runtime cost.

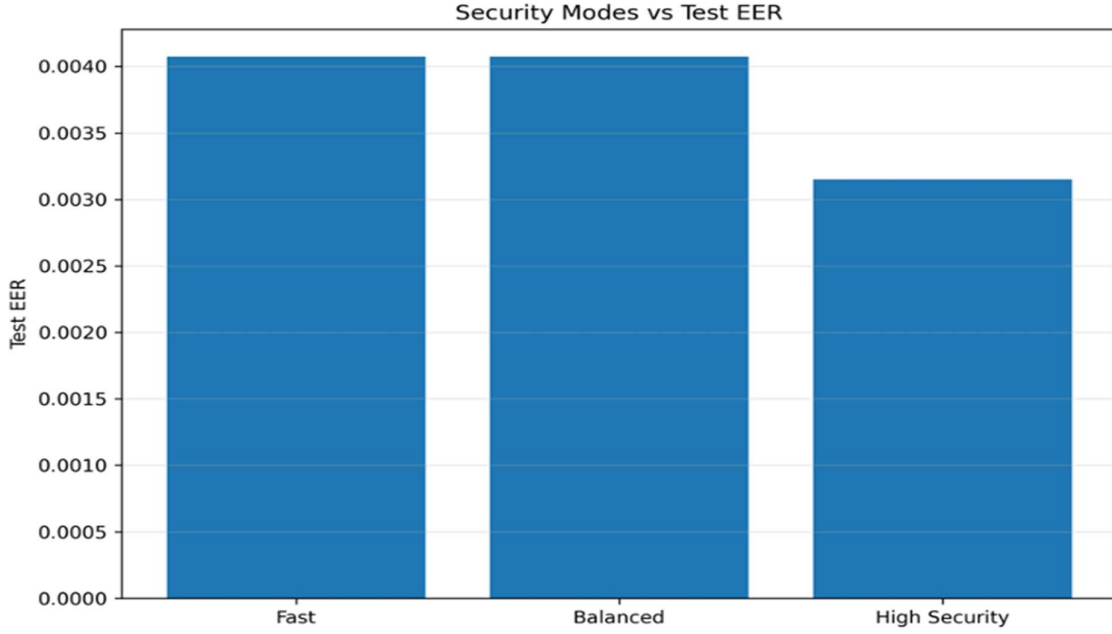


Fig. 6.7. Latency-over-security comparison for protected and protected CKKS matching modes.

6.10 Security–Utility Trade-off Discussion

The main system-level insight from the results is that privacy-preserving multimodal biometric verification cannot be evaluated by a single metric such as EER. A low EER is useful only when the system also supports template renewability, reduced cross-key linkability, encrypted computation, and feasible runtime. Therefore, the proposed framework interprets performance through a combined security–utility view.

The security–utility trade-off in the proposed framework arises because cancellable transformation modifies the original biometric embedding space. This modification is necessary for template renewability and unlinkability, but it may also disturb the similarity relationships required for accurate verification.

If the transformation is too strong, useful identity information may be lost, increasing EER and reducing genuine-user acceptability. If the transformation is too weak, verification performance may remain high, but the protected template may provide weaker unlinkability. Therefore, the

best operating point is not selected only by the lowest EER. It must be selected by considering same-key utility, cross-key unlinkability, fusion recovery, CKKS preservation, and runtime.

The results show that the trade-off is modality-dependent. Iris retains strong protected-domain performance, which indicates that its embedding structure remains discriminative after transformation. Face shows moderate degradation but continues to provide useful complementary evidence. Fingerprint is more sensitive to transformation, but it can still contribute to multimodal fusion.

The same-key and cross-key results provide two complementary views of protection. Same-key results show whether the protected templates remain useful for verification. Cross-key results show whether renewed templates become difficult to link. Protected fusion then shows whether the system can recover performance after protection. CKKS matching shows whether protected similarity computation can be performed privately without changing verification behaviour.

Therefore, the proposed system should be evaluated as a complete security–utility framework. The results establish that cancellable transformation, multimodal fusion, and CKKS encrypted matching should not be evaluated in isolation. Cancellable transformation provides renewability and practical unlinkability, fusion recovers lost utility, and CKKS adds computation-level privacy with runtime overhead rather than recognition loss.

6.11 Overall Findings

The main findings of the experimental analysis are as follows:

1. CKKS encrypted matching preserves plain-domain verification behaviour in the baseline system. This shows that encrypted similarity computation does not materially reduce recognition performance under the tested setting [14], [15].

2. Cancellable transformation introduces a modality-dependent security–utility trade-off. Iris remains highly robust after protection, face shows moderate degradation, and fingerprint is more sensitive to transformation.
3. Same-key evaluation confirms that protected templates can retain verification utility, while cross-key evaluation provides empirical evidence of practical unlinkability and renewability [5], [6].
4. Protected multimodal fusion significantly improves performance over protected unimodal systems. This confirms that fusion works as a utility-recovery mechanism after template protection [7]–[10].
5. Protected CKKS matching preserves protected plain-domain verification behaviour. This indicates that the main cost of CKKS is computational overhead rather than recognition degradation.
6. The trimodal protected system provides the strongest system-level result, achieving low EER while supporting template protection, fusion-based recovery, and encrypted matching.

These findings support the main research contribution of the work: a system-level security–utility framework for privacy-preserving multimodal biometric verification.

6.12 Summary

This chapter presented the results and analysis of the proposed privacy-preserving multimodal biometric verification framework. The baseline results showed that iris achieved the strongest unimodal performance, followed by face and fingerprint. CKKS encrypted matching preserved plain-domain verification behaviour.

The protected unimodal results showed that cancellable transformation affects each modality differently. Iris retained the strongest protected-domain utility, face remained useful with moderate degradation, and fingerprint was more sensitive to transformation. Same-key

evaluation measured retained utility, while cross-key evaluation provided empirical evidence of practical unlinkability and renewability.

Protected multimodal fusion significantly improved verification performance after template protection. The trimodal protected system achieved the strongest performance and provided the highest recovery gain over the best protected unimodal system. Protected CKKS matching preserved protected-domain verification behaviour, showing that encrypted computation introduces runtime overhead rather than recognition degradation.

Overall, the results demonstrate that secure multimodal biometric verification should be evaluated as a layered system. Template protection, unlinkability, fusion recovery, encrypted computation, and runtime cost must be considered together. The next chapter concludes the report by summarizing the major findings, contributions, limitations, and future research directions.

CHAPTER 7

Conclusion And Future Work

7.1 Conclusion

This Major Project-II presented a privacy-preserving multimodal biometric verification framework using face, fingerprint, and iris modalities. The work extended the Major Project-I baseline framework by adding cancellable biometric template transformation, same-key and cross-key evaluation, protected score-level fusion, and CKKS encrypted matching over protected templates.

The main purpose of this work was not only to improve biometric verification accuracy. The central aim was to analyse secure multimodal biometric verification from a system-level security–utility perspective. In biometric systems, accuracy alone is not sufficient because biometric templates are permanent, sensitive, and difficult to revoke after compromise. Therefore, a secure biometric system must also consider template renewability, unlinkability, computation-level privacy, and practical runtime overhead [2]–[6].

The proposed framework addresses these requirements through a layered design. Cancellable transformation provides template-level protection by converting original biometric embeddings into key-dependent protected templates. Same-key evaluation measures whether protected templates retain verification utility. Cross-key evaluation analyses whether templates generated under different keys become difficult to link, thereby supporting practical unlinkability and renewability. Protected score-level fusion evaluates whether multimodal fusion can recover the utility loss introduced by cancellable transformation. CKKS encrypted matching adds computation-level privacy by enabling protected-domain similarity computation without materially changing verification behaviour [5], [6], [11]–[15].

The results show that the proposed framework provides a meaningful balance between recognition performance and privacy protection. The baseline experiments showed that CKKS

encrypted matching preserved plain-domain verification behaviour. After cancellable transformation, protected unimodal performance became modality-dependent. Iris retained the strongest protected utility, face showed moderate degradation, and fingerprint was more sensitive to transformation. This confirms that protected-template configuration should be selected with modality-specific behaviour in mind.

The protected fusion results established an important system-level insight. Multimodal fusion after template protection is not only an accuracy-enhancement step. It acts as a utility-recovery mechanism because different biometric modalities preserve different portions of discriminative identity information after transformation. Since the errors of the protected modalities are not fully identical, score-level fusion can reduce the effect of degradation in any single modality and recover a significant part of the lost utility [7]–[10].

The protected CKKS results further showed that encrypted matching preserves protected-domain verification behaviour. This means that the main recognition loss is due to cancellable transformation, while CKKS mainly contributes computational overhead. Therefore, the two privacy mechanisms play different roles: cancellable biometrics provides template-level renewability and unlinkability, while CKKS provides computation-level privacy [11]–[15].

Overall, the work demonstrates that secure multimodal biometric verification should be evaluated as a layered system rather than as separate template protection, fusion, and encryption modules. The final performance depends on the interaction between protection strength, modality-specific degradation, fusion-based recovery, cross-key unlinkability, encrypted matching preservation, and runtime cost.

7.2 Major Findings

The major findings of this project are summarized as follows.

1. **CKKS preserved baseline verification behaviour:** In the Major Project-I baseline, CKKS encrypted matching produced the same verification behaviour as plain-domain

matching. This indicates that encrypted inner-product computation can preserve recognition performance when it follows the same similarity structure as plain matching [14], [15].

2. **Cancellable transformation introduced modality-dependent effects:** The protected unimodal results showed that face, fingerprint, and iris do not respond equally to cancellable transformation. Iris retained the strongest protected performance, face remained effective with moderate degradation, and fingerprint was more sensitive to transformation.
3. **Same-key evaluation confirmed protected-template utility:** Same-key matching showed that protected templates can retain sufficient discriminative information for verification. This confirms that cancellable transformation does not necessarily destroy matching capability when suitable parameters are selected.
4. **Cross-key evaluation supported practical unlinkability:** Cross-key matching showed that templates generated under different transformation keys become difficult to associate. This provides empirical evidence of practical unlinkability and renewability under the tested protocol [5], [6]. However, this is not claimed as a formal proof of irreversibility.
5. **Protected fusion recovered utility after template protection:** Protected multimodal fusion significantly improved performance compared with protected unimodal systems. The trimodal protected system achieved the strongest result because it combined complementary protected evidence from face, fingerprint, and iris [7]–[10].
6. **CKKS added computation-level privacy without recognition loss:** Protected CKKS matching preserved protected-domain verification behaviour. This shows that CKKS mainly adds runtime overhead rather than additional recognition degradation [14], [15].

7. **Security–utility trade-off must be evaluated jointly:** The results show that a privacy-preserving biometric system should not be selected only by the lowest EER. It should be evaluated using a combined view of same-key utility, cross-key unlinkability, fusion recovery, encrypted computation, and runtime overhead [3]–[6].

7.3 Research Contribution of the Work

The main research contribution of this work is the establishment of a system-level security–utility evaluation framework for privacy-preserving multimodal biometric verification. The work does not claim to introduce a new biometric recognition model, a new cancellable transformation algorithm, or a new cryptographic scheme. Instead, it contributes an analytical framework for studying how existing privacy-preserving mechanisms interact when applied together under one controlled face–fingerprint–iris verification protocol.

The contribution can be understood through four system-level insights.

First, the work shows that cancellable transformation should not be evaluated only by protected unimodal accuracy. A protected template may retain good same-key verification performance, but it must also be examined under cross-key conditions to evaluate practical unlinkability and renewability. Therefore, same-key and cross-key evaluations must be treated as complementary parts of biometric template protection analysis [5], [6], [11]–[13].

Second, the work establishes that protected-template behaviour is modality-dependent. Iris, face, and fingerprint embeddings respond differently to cancellable transformation. This means that a single protection setting should not be assumed optimal for all biometric traits. Modality-aware analysis is necessary for selecting practical protected-template configurations.

Third, the work demonstrates that protected multimodal fusion can act as a utility-recovery layer after template protection. The improvement occurs because different protected modalities preserve different portions of identity-discriminative information. Since their errors are

partially non-overlapping, score-level fusion can recover performance that is lost in individual protected matchers [7]–[10].

Fourth, the work separates the role of template-level protection and computation-level privacy. Cancellable transformation modifies the template space and supports renewability and unlinkability. CKKS encrypted matching preserves protected-domain similarity computation and mainly introduces runtime overhead. Therefore, cancellable biometrics and CKKS should be treated as complementary privacy layers rather than interchangeable security mechanisms [11]–[15].

Thus, the research value of this project lies in showing how template protection, unlinkability evaluation, multimodal fusion, encrypted matching, and runtime overhead should be jointly analysed for secure biometric verification.

7.4 Security–Utility Interpretation

The results of this project show that privacy-preserving biometric verification involves a trade-off between protection strength and recognition utility. Cancellable transformation improves template renewability and reduces cross-key linkability, but it may also disturb the original embedding structure and reduce matching performance [11]–[13].

This trade-off is not uniform across modalities. Iris retained high protected-domain performance because its embedding representation remained strongly discriminative after transformation. Face showed moderate degradation but continued to provide useful complementary evidence. Fingerprint was more sensitive to transformation, but it still contributed auxiliary information in multimodal fusion.

The same-key and cross-key experiments provide two different views of the protected system. Same-key evaluation measures whether the protected representation is useful for verification. Cross-key evaluation measures whether renewed protected templates become difficult to link. A secure cancellable biometric system should satisfy both conditions. Good same-key

performance alone is not sufficient if templates remain linkable across keys. Similarly, strong cross-key unlinkability is not useful if same-key verification becomes too weak [5], [6].

Protected fusion changes the security–utility balance by recovering part of the performance loss introduced by transformation. This means that multimodal fusion is not only a recognition-improvement tool but also an important design component in privacy-preserving biometric systems [7]–[10]. CKKS encrypted matching then adds computation-level privacy while preserving protected-domain verification behaviour [14], [15].

Therefore, the best operating configuration should be selected by considering the complete system: protected utility, cross-key unlinkability, fusion recovery, encrypted computation, and runtime overhead.

7.5 Limitations of the Present Work

Although the proposed framework provides a structured system-level evaluation, the present work has some limitations.

First, the multimodal dataset is virtually subject-aligned. Face, fingerprint, and iris samples are taken from separate public datasets and arranged under common virtual identity labels. This allows controlled multimodal evaluation, but it does not fully represent a naturally collected multimodal dataset where all traits come from the same real individuals.

Second, the work focuses on verification, not large-scale identification. The conclusions are therefore specific to pairwise biometric verification and may require further study for identification scenarios involving large galleries.

Third, the cross-key evaluation provides empirical evidence of practical unlinkability under the tested protocol, but it is not a formal mathematical proof of irreversibility or unconditional unlinkability. A stronger security proof would require additional theoretical and adversarial analysis [5], [6].

Fourth, the work does not address complete secure key management. The transformation key is assumed to be protected and not intentionally disclosed. In real deployment, key generation, storage, rotation, and recovery would require a secure key-management protocol.

Fifth, the threat model assumes an honest-but-curious server. The present work does not analyse malicious servers that may deviate from the protocol, nor does it evaluate side-channel attacks, adversarial attacks, template inversion attacks, or hill-climbing attacks.

Sixth, CKKS encrypted matching introduces computational overhead. Although encrypted matching preserves verification behaviour, its runtime cost must be carefully considered for real-time deployment [14], [15].

These limitations define the boundary of the present study and provide directions for future improvement.

7.6 Deployment Implications

The results of this project suggest that privacy-preserving biometric systems should be deployed according to the required balance between speed, security, and recognition performance.

For applications where speed is the main requirement, protected plain matching may be suitable because it provides cancellable template protection and renewability with lower computational overhead. For applications requiring stronger computation-level privacy, protected CKKS matching is more appropriate because it enables encrypted similarity computation. For high-security applications, trimodal protected CKKS matching provides the strongest layered protection and the best protected verification performance, although it introduces higher runtime cost.

The deployment choice should therefore depend on the application scenario. Low-risk or resource-constrained systems may prefer faster protected matching. High-security systems such as border control, secure access, and sensitive identity management may justify the

additional cost of protected CKKS matching. This shows that the proposed framework can support different operating modes depending on the required security–utility balance.

7.7 Future Work

Several directions can be explored in future work.

First, the framework can be evaluated on a naturally aligned multimodal biometric dataset where face, fingerprint, and iris samples are collected from the same real subjects. This would strengthen the external validity of the results.

Second, a formal irreversibility and unlinkability analysis can be added. While the present work provides empirical cross-key evidence of practical unlinkability, future work can include stronger mathematical or adversarial evaluation [5], [6].

Third, secure key-management mechanisms can be integrated with the cancellable transformation stage. This may include key rotation, user-specific key handling, secure key storage, and recovery procedures.

Fourth, fully encrypted matching can be explored where both templates or additional parts of the fusion process are encrypted. This would provide a stronger privacy-preserving computation model, although it may increase computational cost [14], [15].

Fifth, adversarial robustness can be evaluated. Future experiments may include template inversion attacks, hill-climbing attacks, score leakage analysis, and attacks against protected embeddings.

Sixth, adaptive fusion strategies can be explored. Instead of fixed validation-selected weights, future work may evaluate quality-aware, modality-aware, or security-aware fusion methods that dynamically adjust weights based on the reliability of each protected modality.

Seventh, the framework can be extended to larger datasets and additional biometric modalities such as palmprint, voice, or periocular biometrics. This would help study the scalability of the proposed security–utility framework.

Finally, runtime optimization can be studied for practical deployment. This may include CKKS parameter tuning, batching, vector packing, GPU-supported preprocessing, and approximate matching strategies to reduce encrypted matching latency.

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